The Imagination of Nature:
Mythology as the Cultural Recording Layer of Perceptual Coupling

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Abstract

Does a people's mythology carry a recoverable trace of the biome it grew up in, or is its symbolic content a matter of mind and culture alone? Mythography has mostly treated the imagery of myth as a product of mind and culture, with the environment supplying raw material but not form; a counter-tradition treats imagination instead as a coupling between perceiver and place, on which a mythology should bear the mark of its ecology. We test this with Berezkin's catalogue of folk-motifs from 958 traditions, each geocoded to one of the 14 WWF terrestrial biomes, embedding every motif's text with sentence-pooled SigLIP-2 and aligning it to imagery of its biome (iNaturalist species photographs and Places365 landscape scenes). A tradition's mythology aligns with its own biome's imagery, and this alignment is not reducible to the species, places, or peoples a myth happens to name: it holds whether that identity vocabulary is removed from the text before embedding, held constant under a matched-permutation null, or read off the unsupervised myth-image geometry without biome labels at all, three convergent analytical strategies that agree on the same biomes. The alignment decays monotonically with motif breadth, strongest in biome-specific motifs and fading toward zero in cross-culturally universal ones; this breadth gradient is the load-bearing signature. The trace a mythology carries of its biome lives in the schemas of attention, activity, and form that a place makes available to imagination: the creatures a landscape rewards attention to, the activities it affords, and the shapes that recur in it until they settle into narrative. We read mythology as a cultural recording layer of this coupling between people and place, the medium through which a tradition keeps its long engagement with its environment legible across generations. A computational mythology that can now read such records at scale, tracing how motifs travel between cultures and shift through time, could test directly whether this coupling stabilised over generations, as the present snapshot suggests.

Significance Statement

Mythologies of cultures inhabiting different biomes carry a recoverable visual signature of those biomes, even after every species name, place name, and biome word is stripped from the text. The signature is strongest in biome-specific myths and attenuates progressively as motifs propagate across more biomes. The pattern is consistent with the hypothesis that the ecological structure of a biome is one of the factors that stabilise its mythological motifs, and gives empirical purchase on a question that recent theories of culture recast not as *whether* imagination projects meaning onto a passive world or couples to an articulate one, but as *how much* the environment weights the forms a mythology settles into.

Introduction

Modern thought has long treated nature as a passive backdrop onto which the human mind projects meaning. Mythology, on this view, is what humans *do to* a world, a projection of cultural categories, anxieties, and social structures onto matter. The biome a culture inhabits supplies raw material but not, on this reading, the *form* of its symbols. We call this dominant view the *projection account*. Running parallel to this default, however, a different tradition has insisted on a different picture. Schelling's distinction between *natura naturata* (nature as finished product) and *natura naturans* (nature as productive activity) [Schelling, 1797, 1799] was recovered in the twentieth century by Bateson's claim that "mind is wherever the conditions for a mental process obtain" across coupled systems, where differences make differences and pattern propagates between brain, body, environment, and culture [Bateson, 1972, 1979]. Depth psychology made a related move. Jung's mature view of the archetypes resisted easy translation into innate mental contents stored in individual brains and instead located them in the joint between psyche and nature [Jung, 1959]; Hillman radicalised this into archetypal psychology and the recovery of *anima mundi* [Hillman, 1975, 1981]; Henri Corbin gave it its most precise philosophical name in the *mundus imaginalis*, the imaginal world, distinct from the modern collapse of "imaginary" into "merely subjective" [Corbin, 1972, Barfield, 1957, Abram, 1996, Ingold, 2000]. The thread connecting these authors is that imagination is a perceptual organ rather than a productive one, an organ that discloses content present in the encounter between an attentive perceiver and an articulate world. We call this second view the *coupling account*, since on this reading perceiver and environment co-produce symbolic content together.

Since the 1960s the cognitive sciences have been producing the empirical scaffolding for exactly this coupling view, on a philosophical footing Merleau-Ponty had already laid: perceiving is not a spectator's representation of a world set apart, but the engagement of a body always already situated in it [Merleau-Ponty, 1945]. Gibson's ecological psychology replaced the dominant representational model of perception with one of *direct pickup*, in which the environment is information-rich at the level of the optic array and the organism tunes itself to a richly structured world that already contains the relevant information [Gibson, 1979]. The 4E framework (embodied, embedded, extended, and enactive cognition) generalised this beyond perception, drawing Merleau-Ponty's phenomenology into cognitive science explicitly [Varela et al., 1991, Clark and Chalmers, 1998, Noë, 2004]. Predictive processing supplied the neural substrate, with the brain treated as a hierarchical prediction machine that locks onto interpretations its priors privilege, given the actual statistical structure of what the world presents [Friston, 2010, Clark, 2016, Hohwy, 2013]. On this account pareidolia is not a mistake of perception but the system working as designed. Niche construction theory closed the loop [Odling-Smee et al., 2003], observing that humans produce symbolic environments (names, stories, transmitted practices) and that these environments shape the cognition of subsequent generations.

These threads converge, in more recent work, on a picture that supersedes the projection-versus-coupling choice rather than taking a side in it. Cultural attraction theory treats the stability of a cultural form not as a victory of mind over matter or matter over mind, but as its settling at an *attractor* fixed jointly by cognitive factors of attraction — what minds find memorable, inferable, and transmissible — and *ecological* factors of attraction, the material and environmental conditions under which the form is repeatedly produced and reproduced [Sperber, 1996, Claidière et al., 2014, Morin, 2016]. Cognitive ecology names the same web of mutual dependence among brain, body, environment, and culture [Hutchins, 2010]; the cultural-affordances programme casts a lived environment as a structured landscape that scaffolds attention and shared sense-making [Ramstead et al., 2016]; and niche-construction accounts formalise the loop by which the worlds people build come to shape the cognition of those who inherit them [Constant et al., 2018]. On this integrative view, projection and coupling are not rival theories of what imagination *is*, but name the two ends of a single, measurable dimension: how heavily the ecological factors of attraction weight the attractors at which a tradition's mythic forms come to rest.

This dimension is empirically tractable, and its projection end is a sharp null. In that limit ecological factors of attraction carry no weight and mythic form is fixed entirely by cross-cultural cognitive and social attractors, so once a mythology's text has been stripped of its overt environmental and cultural anchors (species names, place names, ethnonyms, biome words, climate phrases, tradition labels), the residual narrative should carry no detectable visual signature of the biome the tradition inhabits — only a cross-cultural grammar of plot and relation, indifferent to where it emerged. Any ecological weight, by contrast, should leave a recoverable residue in the surviving text, because that residue was part of what stabilised the mythology in the first place; and a graded form of the same expectation is that the residue should be strongest in motifs whose distribution stayed close to a single biome and fade as motifs propagate across many, since widely-shared motifs are precisely those whose stabilising encounters were not tied to one environment. Measuring whether such a residue exists, and how it scales with breadth, estimates where a mythology's forms sit on the dimension — without committing to the strongest reading, on which imagination becomes a perceptual organ; that stronger interpretation, and its limits, we take up in the Discussion.

Two recent developments make this question newly testable at scale. The first is the cross-cultural folk-motif catalogue assembled by Yu. E. Berezkin, comprising 2,564 motifs across 958 traditions with Russian-language abstracts and geocoded centroids [Berezkin, 2024], which we have joined to the WWF Ecoregions classification of 14 terrestrial biomes [Olson et al., 2001]. The second is the development of frontier vision–language models such as SigLIP-2 [Tschannen et al., 2025], which extends the sigmoid image–text objective of SigLIP [Zhai et al., 2023] and embeds text and image in a shared metric space learned from billions of image–text pairs sampled from the internet. The combination lets us ask whether, after stripping every species name, every place name, every biome word, and every tradition label, the embedding of a tradition's mythology preferentially aligns with images of its own biome, and whether that alignment is concentrated in biome-specific motifs in the way an ecological-affordance reading would predict.

We report below that such a residue is detectable, and that it survives three independent tests whose assumptions do not overlap (Figure 1): removing the identity names from the text before embedding, holding them statistically constant under a matched-permutation null on the raw myths, and recovering biome structure from the unsupervised myth–image geometry without ever using biome labels. The three converge on the same biomes. After stripping every overt environmental and cultural anchor from the text, the embedding of a tradition's mythology preferentially aligns with images of its own biome on two independent image corpora and, on the iNaturalist species channel, across four vision–language model architectures; the matched-permutation control reproduces the alignment on the original, un-anonymised Russian text. The alignment is graded with motif breadth in the way an ecological-affordance reading would predict, strongest in motifs that did not propagate beyond a single biome and attenuating in motifs shared across many. We take this pattern to be more naturally accommodated by a coupling or ecological-affordance account than by a view in which biome supplies only interchangeable narrative material, and read mythology as the surviving textual residue of an ongoing perceptual coupling between cognition and environment, the recoverable trace in transmitted narrative of a living engagement that the narrative itself does not exhaust.

Results

Biome-specific mythology aligns with biome imagery

The Berezkin corpus distributes 958 traditions across the 14 WWF biomes (Figure 2). After the full anonymisation pipeline described in *Materials and Methods*, the LLM-mediated translation and double-pass refinement that replaces every species with a class word, every place and ethnonym with a place-holder, and every tradition-specific character name and cultural-religious title with a neutral generic, the residualised alignment on the full LLM-clean corpus (2,158 motifs, sentence-pooled SigLIP-2) reaches within-iconic-taxon stratified $\mu\Delta = +0.406 \times 10^{-3}$ across the 14 biomes on iNaturalist (Figure 3, Panel A; 8/14 biomes significant at $p < 0.05$). The strongest positive cells under stratification lie in Montane Grasslands & Shrublands ($\Delta^{\text{strat}} = +1.08 \times 10^{-3}$, $p < .001$), Tropical & Subtropical Moist Broadleaf Forests ($+0.93 \times 10^{-3}$, $p < .001$), Tropical & Subtropical Grasslands, Savannas & Shrublands ($+0.76 \times 10^{-3}$, $p < .001$), Boreal Forests/Taiga ($+0.65 \times 10^{-3}$, $p < .001$), Deserts & Xeric Shrublands ($+0.63 \times 10^{-3}$, $p < .001$), Tropical & Subtropical Dry Broadleaf Forests ($+0.44 \times 10^{-3}$, $p < .01$), Tundra ($+0.36 \times 10^{-3}$, $p < .01$), and Temperate Grasslands, Savannas & Shrublands ($+0.33 \times 10^{-3}$, $p < .01$). The remaining biomes are positive or near zero with the exception of Mediterranean Forests

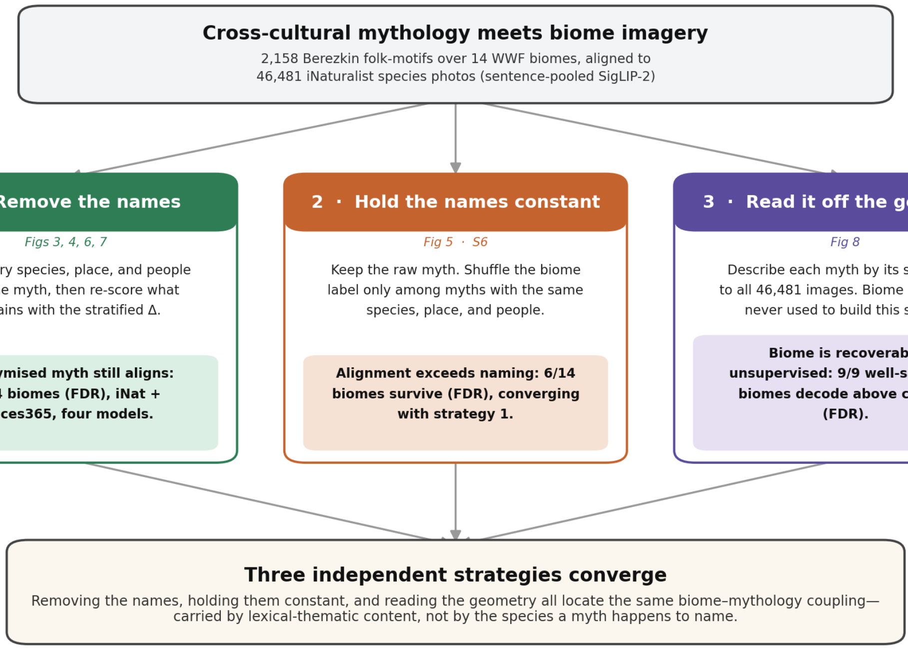

Figure 1. Three independent strategies for asking whether mythology carries a trace of its biome. We test the same question three ways whose assumptions do not overlap. *(1) Remove the names*: anonymise the text (every species $\to$ its class word, place and ethnonym $\to$ a placeholder, biome words dropped) and measure the within-iconic-taxon stratified alignment to biome imagery. *(2) Hold the names constant*: keep the raw myths intact and break the biome–identity correlation with a matched-permutation null that shuffles biome only among motifs of like species, place, and ethnonym content. *(3) Read it off the geometry*: describe each myth by its similarity profile across the whole image set and recover biome structure without ever using biome labels to build the representation. The three strategies converge on the same biome–mythology coupling, carried by lexical-thematic content rather than by the species a myth happens to name. The figure pointers index the Results subsections that follow.

and Temperate Conifer Forests, both weakly negative under stratification.

The biomes that survive the within-iconic-taxon control therefore carry alignment that is not attributable to biome-by-iconic-taxon co-occurrence on the image side; the cells where stratification narrows the marginal effect precisely identify where the marginal signal was riding on taxonomic composition rather than on biome-character imagery. Mediterranean Forests is the cleanest example: its marginal $\Delta = +0.51 \times 10^{-3}$ ($p < .001$) is carried by a single strongly positive Plantae cell ($+1.38 \times 10^{-3}$ under the per-taxon decomposition, Figure 7), because Mediterranean iNaturalist imagery is dominated by plants (vineyards, olive groves, shrubland) and the motifs assigned to Mediterranean biomes lean similarly plant-coded. Once we hold iconic-taxon fixed on the image side, the Plantae contribution is uniformly averaged against the negative Aves (-0.81), Animalia (-1.12), Reptilia (-0.39), and Amphibia (-0.91) cells, and the stratified Δ collapses to -0.09×10^{-3} (panel A bottom). The reverse case is also visible. Boreal Forests/Taiga has a near-zero marginal Δ ($+0.03 \times 10^{-3}$) because its dominant Plantae stratum (conifer imagery) carries no distinctive boreal mythological content; once stratification gives Mammalia (bears, wolves, $+1.77 \times 10^{-3}$) and Actinopterygii ($+1.39$) equal weight, the boreal Δ emerges at $+0.65 \times 10^{-3}$ ($p < .001$).

The same pattern replicates on an independent image corpus. The Places365 strict-landscape comparison set, 1,655 scene photographs across 11 biomes drawn from Places365 scene categories that unambiguously match a single WWF biome and then SigLIP-filtered against humans, vehicles, buildings, and close-up animal subjects, gives a marginal $\mu\Delta = +0.423 \times 10^{-3}$ across 11 biomes, with 6 biomes significant at $p < 0.05$ (Figure 3, Panel B). The strongest cells fall in Tropical & Subtropical Moist Broadleaf Forests ($\Delta = +1.67 \times 10^{-3}$, $p < .001$), Mediterranean Forests, Woodlands & Scrub ($+1.25 \times 10^{-3}$, $p < .001$), Boreal Forests/Taiga ($+1.06 \times 10^{-3}$, $p < .001$), Deserts & Xeric Shrublands ($+0.94 \times 10^{-3}$,

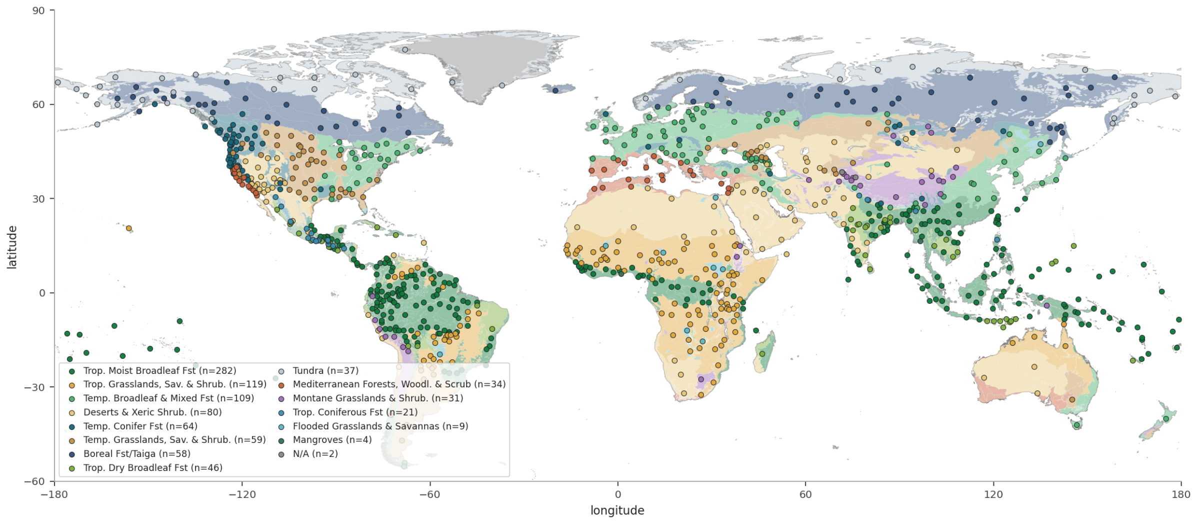

Figure 2. Cross-cultural sample of mythology over the world's biome geography. World map of the 958 Berezkin traditions plotted as dots over the underlying WWF Ecoregions polygons, both coloured by the same biome palette: continents in semi-transparent biome fills (the Saharan and Australian deserts in tan, the Amazonian and Indo-Malay rainforests in dark green, the Boreal belt across northern Eurasia and Canada in muted blue, Mediterranean basins in red, etc.), tradition centroids as opaque dots in the same palette. The overlay shows how densely each biome's actual extent is sampled by Berezkin's corpus. Eurasian and North American traditions are relatively dense; some biomes (Mangroves, Flooded Grasslands and Savannas) are sparsely covered.

$p < .001$), and Montane Grasslands & Shrublands ($+0.57 \times 10^{-3}$, $p < .001$). Crucially, the two corpora cover different visual content: iNaturalist samples close to species-level (foliage, fauna, fungi in their habitat), Places365 samples wide-shot terrain with no species or human subjects. The fact that anonymised mythology aligns above permutation chance with both channels of biome imagery, species-level and scene-level, means the alignment is not an artefact of one corpus's curatorial peculiarities. The per-biome pattern is likewise not an artefact of a single encoder: it correlates across four vision–language models (SigLIP-2, M-CLIP, and OpenCLIP under LAION-2B and OpenAI pretrainings), with across-model Pearson correlations of the per-biome Δ in the range $r = 0.74$–0.94, though the two OpenCLIP variants carry a weaker overall mean (Supplementary Section S3, Figure 13). Projected onto a world map (Figure 4), the warmest cells concentrate in the Amazonian and Indo-Malay Tropical Moist Broadleaf rainforests, the Saharan and Australian deserts, the Mediterranean basins of southern Europe and the western Maghreb, the boreal belt across northern Eurasia and Canada, and the montane grassland systems of the high-altitude tropics.

Cultural-geographic autocorrelation

Mythologies, traditions, and biomes are spatially and historically clustered. A biome effect could in principle reflect macroarea, language family, colonial documentation history, or cultural phylogeny rather than ecological coupling. The difficulty is sharpest at long range: using whole-genome data to ask how far folktale distributions track population movement, Bortolini et al. [2017] found genetic and geographic relatedness effectively inseparable at cross-continental scales, and cautioned against reading long-range cultural transmission off extant genetic variability. That is an argument for testing biome within continental blocks rather than across them, which is what we do here. To probe this we ran a within-Glottolog-macroarea biome-swap null. Each Berezkin tradition was assigned to one of the six Glottolog macro-areas [Hammarström et al., 2024] (Africa, Australia, Eurasia, North America, Papunesia, South America) based on its centroid coordinates. For each of 1,000 permutations, every tradition's motifs were randomly reassigned to a *different* biome from among those represented in its own macroarea: the tradition's macro-area was preserved and its biome label shuffled, so within-macroarea cultural autocorrelation and the colonial documentation history that tracks it remained intact while biome assign-

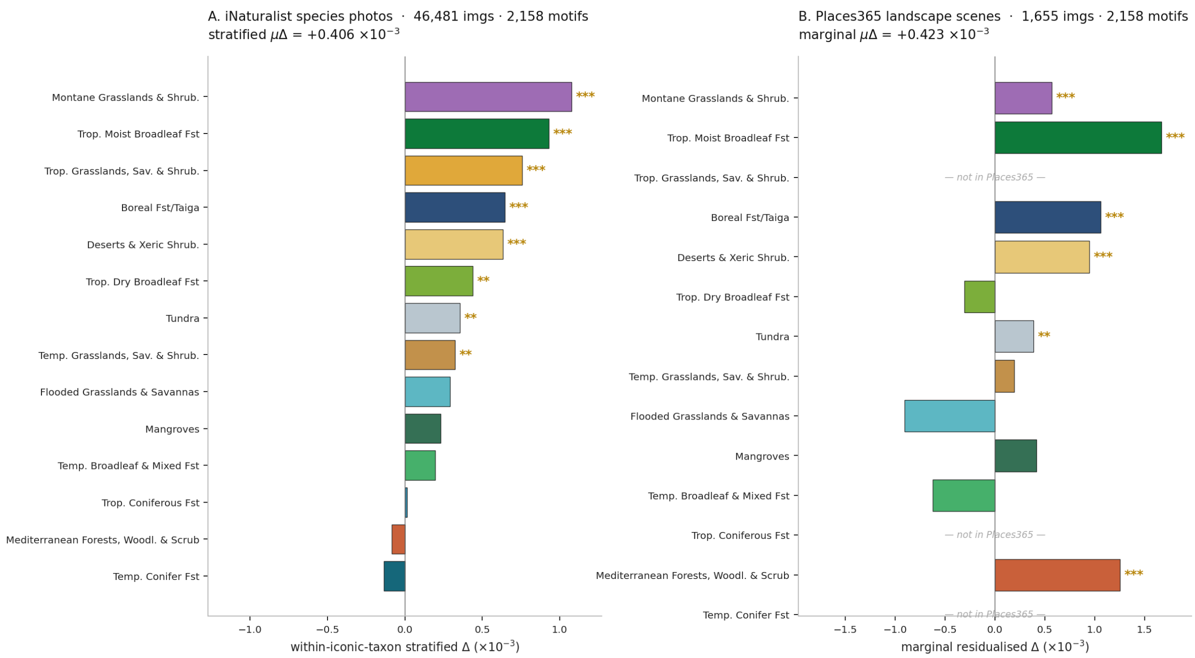

Figure 3. The headline coupling signature, replicated on two independent image corpora. Residualised Δ per WWF biome under sentence-pooled SigLIP-2 on the full LLM-clean Berezkin abstracts (2,158 motifs). *Panel A*: iNaturalist species photographs (n = 46,481 images), with Δ stratified within each iconic-taxon stratum and uniformly averaged across taxa to remove biome–by–iconic-taxon co-occurrence as a possible explanation; stratified $\mu\Delta = +0.406 \times 10^{-3}$, 8/14 biomes significant at $p < .05$. *Panel B*: Places365 strict-landscape scenes (n = 1,655 images, 11 biomes); because Places365 has no iconic-taxon labels (it is scene-only imagery), we report the marginal residualised Δ; marginal $\mu\Delta = +0.423 \times 10^{-3}$, 6/11 biomes significant at $p < .05$. Biomes are ordered top-to-bottom by descending iNat stratified Δ; biomes Places365 does not cover are blanked. Each panel uses its own dynamic range. Gold bold asterisks denote significance after Benjamini–Hochberg FDR correction ($q < .05$); grey asterisks would denote biomes nominally significant at $p < .05$ but not surviving FDR, of which there are none here (every marked biome survives FDR, iNat 8/8 and Places365 6/6). The two corpora agree on the biomes that drive the effect (Tropical Moist Broadleaf, Deserts, Boreal/Taiga, Montane Grasslands, Tropical Grasslands), an independent landscape-channel replication of a signal first recovered on species-channel photographs.

ment broke. We then recomputed the within- iconic-taxon stratified Δ under each swap. The cross-biome mean attenuates by roughly half but does not vanish, and six of the 14 biomes retain observed stratified Δ above the 95th percentile of the swap-null distribution at $p < .05$: Montane Grasslands & Shrublands (observed $+1.08 \times 10^{-3}$, swap-null mean $+0.09$, $p < .001$), Tropical & Subtropical Moist Broadleaf Forests ($+0.93$ vs -0.16, $p < .001$), Boreal Forests/Taiga ($+0.65$ vs $+0.08$, $p < .001$), Deserts & Xeric Shrublands ($+0.63$ vs $+0.22$, $p = .018$), Tropical & Subtropical Grasslands & Savannas ($+0.76$ vs $+0.45$, $p = .019$), and Temperate Grasslands, Savannas & Shrublands ($+0.33$ vs $+0.02$, $p = .022$). The full per-biome table and the visualised swap-null distributions are reported in Supplementary Section S4 and Figure 15. We do not separately preserve language family in this null because many language families in Berezkin's catalogue touch only one or two biomes and admit no swap target; macro-area is the level at which the test has statistical purchase across all 14 biomes. Geographic and cultural autocorrelation therefore contribute to the headline effect but do not exhaust it: the biome content of the motif text carries information beyond within-macroarea cultural-region membership. Nor is the alignment reducible to biome-correlated vocabulary the text makes directly predictable: when motifs are split by an independent, image-free classifier that predicts biome from the anonymised text alone (a simple bag-of-words TF–IDF model, evaluated out-of-fold), the half whose biome that classifier cannot guess still aligns at $+0.40 \times 10^{-3}$, essentially the full-corpus headline. The signal is thus carried by lexical–thematic content beyond the biome-diagnostic words a text model can exploit (Supplementary

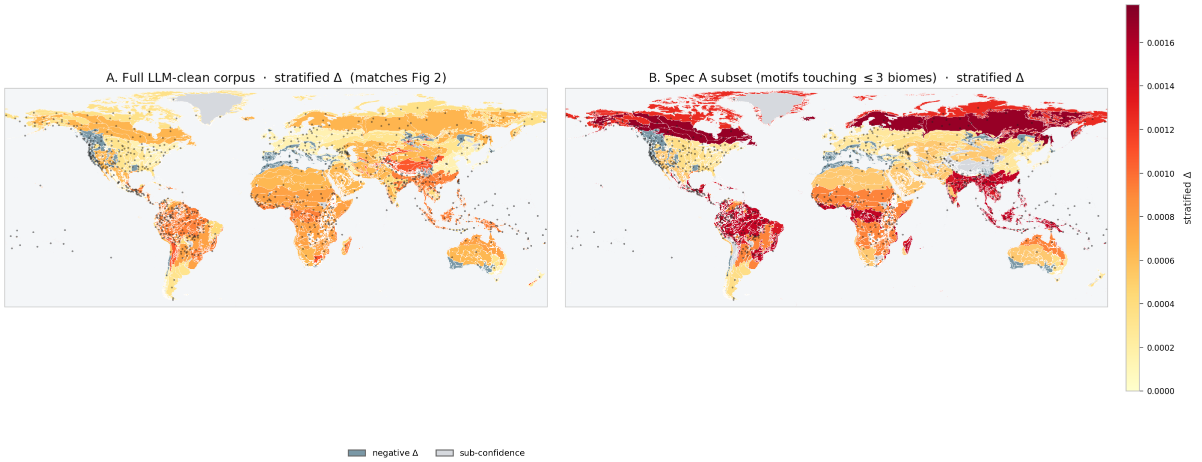

Figure 4. Per-biome Δ projected onto the world map. Geographic companion to Figure 3, on the within-iconic-taxon stratified Δ under sentence-pooled SigLIP-2. *Panel A*: full LLM-clean corpus ($2{,}158$ motifs), matching the headline statistic of Figure 3, Panel A. *Panel B*: Spec A subset (motifs touching ≤ 3 biomes), the breadth-restricted view that the coupling reading predicts should show the cleanest geography (cf. Figure 6). Colour intensity scales with stratified Δ further weighted by confidence $(1-p)$; the YlOrRd colormap is shared between panels so cell intensities are directly comparable. Light grey marks sub-confidence biomes (fewer than 10 traditions, 50 motifs, or 50 photographs); muted slate-blue marks the rare biomes whose stratified Δ falls below zero (in panel A, Mediterranean Forests and Temperate Conifer Forests). Black dots overlay the 958 tradition centroids. The headline coupling geography appears in both panels (Tropical Moist Broadleaf rainforests of the Amazon and Indo-Malay, Tropical Grasslands of sub-Saharan Africa, Boreal Forests/Taiga of northern Eurasia and Canada, Deserts of North Africa and central Australia, Montane Grasslands of the high-altitude tropics) and is substantially more vivid in the Spec A panel where breadth dilution is removed.

Section S4, Figure 14).

The alignment is not reducible to identity naming

The most direct objection to the headline is that a biome's mythology simply names that biome's species, places, and peoples, and the encoder reads the alignment off those names. The anonymisation pipeline answers this by *removing* the names. A complementary strategy keeps the original myths intact and instead *holds the names constant*: a statistical control applied on the same within-iconic-taxon stratified Δ as the headline, so the two are directly comparable. We extract three identity classes per motif from the raw Russian text and keep them separate: the species named in the myth (LLM extraction, adversarial audit 96.7% recall and 95.3% precision), the WWF-codable geographic regions from Berezkin's metadata, and the ethnolinguistic groups that carry the motif. Each is embedded as a bag of its entities with the same sentence-pooled SigLIP-2 pipeline and scored against the same iNaturalist images (Figure 5).

Each identity channel is biome-diagnostic on its own (Figure 5A). A species-only bag gives stratified $\mu\Delta = +0.97 \times 10^{-3}$ ($13/14$ biomes significant), an ethnonym-only bag $+0.73 \times 10^{-3}$ ($14/14$), and a place-only bag $+0.36 \times 10^{-3}$ ($14/14$). The full original myth gives $+0.42 \times 10^{-3}$ ($10/14$), close to the $+0.41 \times 10^{-3}$ the *anonymised* corpus reaches on the same statistic — two opposite roads, removing the names and keeping them, of comparable magnitude (the raw-Russian and English LLM-clean frames are not formally comparable, so we read the closeness as suggestive rather than identical; see *Materials and Methods*). A bag of species names is in fact more biome-diagnostic than the full myth, which dilutes the species cue with narrative content, and this holds under stratification, so it is not an artefact of image-side taxon composition. Species naming is therefore a genuinely strong biome channel, exactly the concern.

We test whether the full-myth alignment exceeds what that naming determines, in two independent ways. A matched-permutation null (Figure 5B) shuffles biome labels only among motifs that are nearest neighbours in the relevant identity-bag embedding (60 K-means blocks), holding identity approximately constant while breaking biome, and recomputes the stratified Δ. Six of the fourteen biomes retain alignment above a joint null that holds species, place, and ethnonym constant simultaneously — Tropical Moist and Tropical Dry Broadleaf Forests, tropical grasslands and savannas, montane grasslands, boreal forest, and mangroves, with five of these surviving every single-channel control as well (Figure 5B). This is the same count as survive the Glottolog cultural-geographic swap, with four of the six biomes shared between the two lists — two stringent and unrelated controls leaving the same handful of biomes standing. (The place channel alone is near-circular, since biome is derived from the region metadata, so we read the joint and species-only nulls rather than place.) An independent embedding-space test (Figure 5C) projects the top species-subspace directions out of the full-myth embeddings and recomputes the stratified Δ, which attenuates from $+0.42$ to $+0.20$–0.27×10^{-3} but stays significant in 8 to 12 of 14 biomes across 5, 10, and 20 removed directions. Neither the matched-permutation null nor the linear projection is an exact control, and both attenuate the effect. What they jointly establish, from assumptions that do not overlap with the anonymisation — the anonymisation route trusts the cleaning but removes the taxon confound, this route trusts no cleaning and, on the stratified statistic, removes it too — is that the alignment on the un-anonymised myths is not reducible to the trivial channel by which a biome's mythology names its own species, places, and peoples. Removing the names and holding them constant are not merely jointly significant in the aggregate: placed side by side, the two strategies recover the same biome geography, the same breadth gradient, and the same per-taxon structure (Spearman $\rho = +0.75$ on the per-biome Δ; Section S6, Figures 17–19).

Breadth gradient

The breadth-stratified analysis in Figure 6 provides the single test that most informatively distinguishes the two accounts. A view in which biome supplies only interchangeable narrative material would expect roughly similar per-biome alignment magnitudes for biome-specific and universal motifs, since both are equally available as cultural projections of biome-correlated style. A view in which local ecological affordances contribute to motif stabilisation would expect biome-specific motifs to be the stable points of encounter with their biome's visual structure, with universals having propagated precisely because they detached from any specific biome and stabilised on cross-cultural structures (family conflict, lost-then-found, transformation). The breadth-split result moves cleanly in the second direction. Marginal $\mu\Delta$ decays from $+0.52 \times 10^{-3}$ on Spec A motifs through $+0.26 \times 10^{-3}$ on semi-universals to $+0.05 \times 10^{-3}$ on universals; stratified $\mu\Delta$ decays from $+0.58 \times 10^{-3}$ through $+0.44 \times 10^{-3}$ to $+0.12 \times 10^{-3}$. Five of fourteen Spec A biome cells are individually significant under within-taxon stratification, against seven of fourteen for semi-universals and two of fourteen for universals; the Spec A subset carries a much larger per-biome effect with proportionally similar significance density. The raw, pre-anonymisation ecological composition of the Spec A motif text per biome — which taxonomic and botanical categories each biome's biome-specific motifs actually name — is given in Supplementary Figure 16. The signature is present where an ecological-affordance reading would locate it, attenuated where breadth dilutes the environmental tie, and essentially absent where motifs have propagated across ≥ 8 biomes.

A text-property alternative could in principle produce this gradient without any ecological decoupling, since universal motifs are longer and more abstract — here breadth correlates with motif length at $r = +0.27$ — so a more diffuse, less biome-lexical text might simply yield a smaller Δ. Recomputing the gradient within motif-length terciles rules this out for the bulk of the corpus: the decline from biome-specific to universal motifs persists within the short and mid-length terciles (short, $+0.59 \rightarrow +0.42 \rightarrow +0.13 \times 10^{-3}$; mid, $+0.94 \rightarrow +0.27 \rightarrow +0.03$), where length is held approximately constant and cannot drive it. The gradient does interact with length, flattening and reversing within the longest tercile ($+0.48 \rightarrow +0.51 \rightarrow +0.60$, where long universal narratives accumulate enough biome-relevant content to align), so it is best read as a property of the shorter, more biome-anchored motifs rather than a uniform law. The length-control script is released with the code.

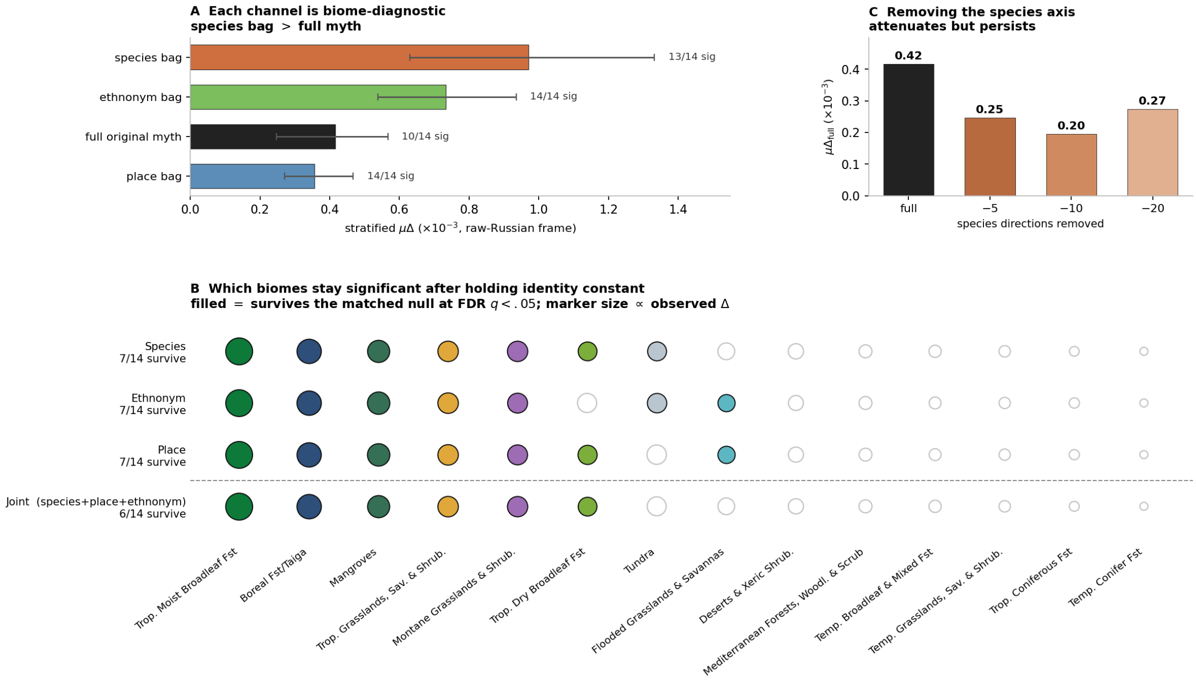

Figure 5. The alignment on the original, un-anonymised myths is not reducible to identity naming. All panels use the within-iconic-taxon stratified Δ in the raw-Russian frame against iNaturalist images. *Panel A*: mean Δ across the 14 biomes (with 95% bootstrap CI and the count of individually significant biomes) for the full original myth and three separately-measured identity baselines (species, place, ethnonym bags). A species-only bag is more biome-diagnostic than the full myth, and this survives stratification. *Panel B*: per-biome significance under each matched-permutation null. For each identity class, and for all three jointly, biome labels are shuffled only among motifs matched on that class's identity-bag embedding (60 K-means blocks); a filled marker means the biome's observed stratified Δ exceeds that null at Benjamini–Hochberg FDR $q < .05$, an open marker that it does not, and marker size is proportional to the observed Δ. Six biomes — Tropical Moist and Tropical Dry Broadleaf Forests, tropical grasslands and savannas, montane grasslands, boreal forest, and mangroves — survive the joint null that holds species, place, and ethnonym constant simultaneously, and five of these survive every single-channel control. The place null is near-circular because biome is derived from the region metadata. *Panel C*: independent embedding-space control — the top k species-subspace directions are projected out of the full-myth embeddings and the stratified Δ recomputed ($k = 5, 10, 20$); $\mu\Delta$ attenuates but stays positive and significant in the majority of biomes.

Per-taxon decomposition

The per-taxon decomposition reports the per-cell *marginal* Δ within each biome–taxon stratum. Because each cell is already conditioned on a single iconic taxon, no further taxon stratification applies within the cell, and the headline within-iconic-taxon stratified Δ on iNaturalist reported in Figure 3 is mechanically the uniform mean of these per-cell marginal values across taxa. The matrix is therefore not a separate analysis from the headline but its building blocks.

Reading the per-biome × per-taxon matrix (Figure 7, Panel A; a per-taxon small-multiples rendering is given in Supplementary Figure 12) row by row reveals that every biome's strongest cells fall in the taxa ecologically characteristic of it. Boreal Forests/Taiga is carried by Mammalia, Aves, and Actinopterygii (bears, owls, salmon); Tundra by Mammalia and Aves; Montane Grasslands by Mammalia, Reptilia, and Insecta; the wet tropics (Tropical Moist Broadleaf, Tropical Grasslands & Savannas) by Reptilia, Amphibia, Plantae, and Fungi; Mediterranean Forests almost exclusively by Plantae; Deserts & Xeric Shrublands by Mammalia, Plantae, and Reptilia. The alignment is not concentrated in any single class but distributes across the per-biome ecologically appropriate fauna and flora.

A trivial reading of this matrix is that the class words themselves are doing the work. Biome b's photos contain biome b's characteristic fauna, biome b's anonymised motif text references those fauna

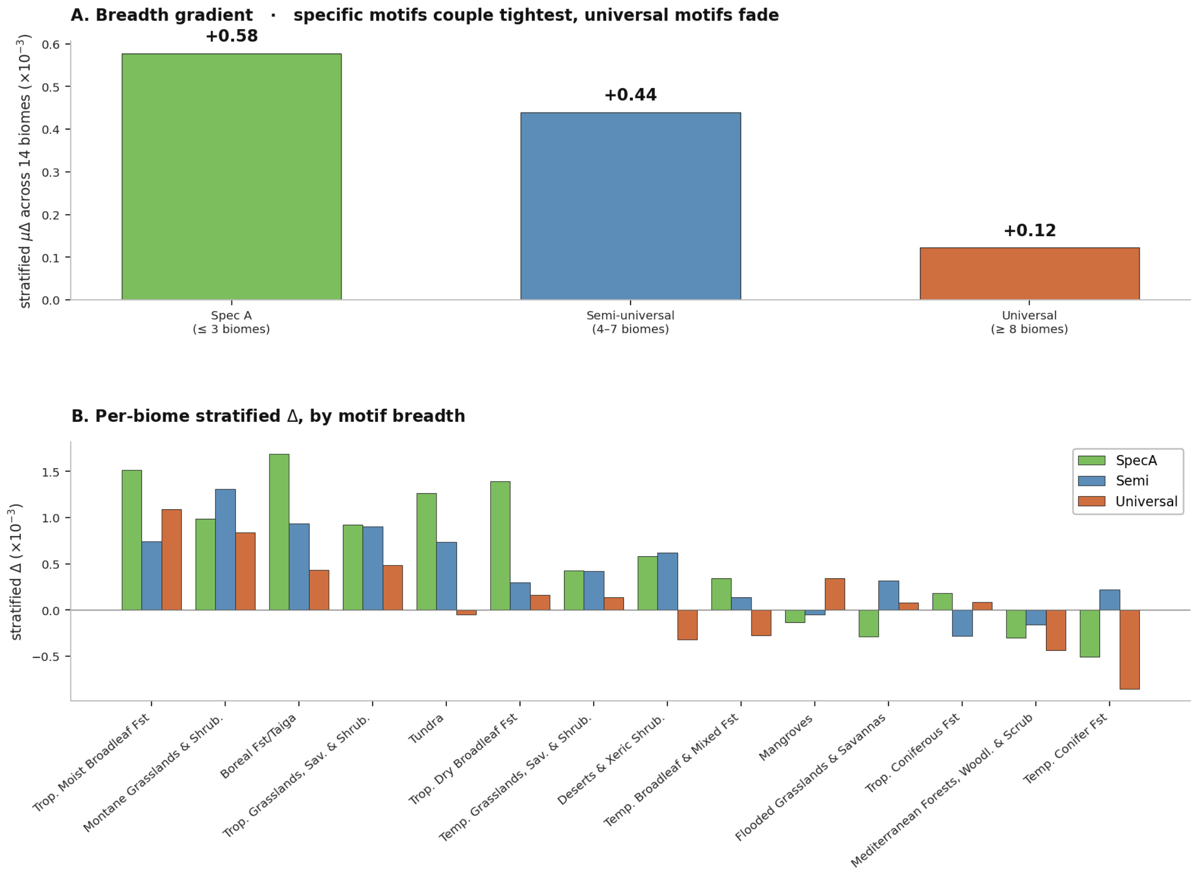

Figure 6. Breadth-stratified analysis. *Panel A*: across-biomes mean within-iconic-taxon stratified Δ on iNaturalist, partitioned by motif breadth into three groups (Spec A: motifs touching $\leq$ 3 biomes; semi-universal: 4–7 biomes; universal: $\geq$ 8 biomes). The stratified $\mu\Delta$ decays monotonically from $+0.58 \times 10^{-3}$ (Spec A) through $+0.44 \times 10^{-3}$ (semi-universal) to $+0.12 \times 10^{-3}$ (universal); the marginal Δ declines in parallel ($+0.52 \rightarrow +0.26 \rightarrow +0.05 \times 10^{-3}$, reported in the text). This is the pattern most naturally expected if local ecological affordances contribute to motif stabilisation. *Panel B*: per-biome stratified Δ within each breadth group, with biomes ordered by mean stratified Δ across the three groups. Spec A bars (green) systematically dominate at the high-Δ biomes (Tropical Moist Broadleaf, Boreal/Taiga, Tropical Grasslands, Tundra, Tropical Dry Broadleaf); universal bars (red) systematically lie at or below zero across most biomes.

by class word ("mammal" more often in boreal mythology, "reptile" more often in tropical mythology) because biome b's mythology grew up around those animals, SigLIP notices the alignment as a class-word-frequency artefact, and the cells are positive by simple co-occurrence. The class-word collapse control in Panel B tests this directly. We re-embedded the same LLM-clean corpus after collapsing every animal-kingdom class word (mammal, bird, fish, reptile, amphibian, insect, mollusc, crustacean, arachnid) to the single token "animal" and every plant-kingdom word (tree, flower, plant, fungus) to "plant". Class-word frequency is now constant across biomes by construction. Every animal mention reads "animal", every plant mention reads "plant", no biome's text carries any class-word information beyond "some animal" or "some plant".

The result is informative on both sides. The pooled alignment attenuates by about a third (the `all` column $\mu\Delta$ drops from $+0.244 \times 10^{-3}$ to $+0.160 \times 10^{-3}$, with the matrix retaining a per-cell Pearson $r = +0.57$ against its class-word-preserved counterpart). Per-taxon means change substantially for the vertebrate animal classes: Mammalia falls from $+0.82$ to $+0.09 \times 10^{-3}$, Reptilia from $+0.74$ to -0.16, Amphibia from $+0.33$ to -0.14. Plant content is more robust: Plantae drops only from $+0.61$ to $+0.48 \times 10^{-3}$. A substantial component of the per-(biome, taxon) signal for animals *was* class-word frequency mirroring image-class frequency. But the pooled effect does not collapse to zero, and the biomes whose cells stay strongly positive under the collapse (Tropical Moist Broadleaf, Temperate

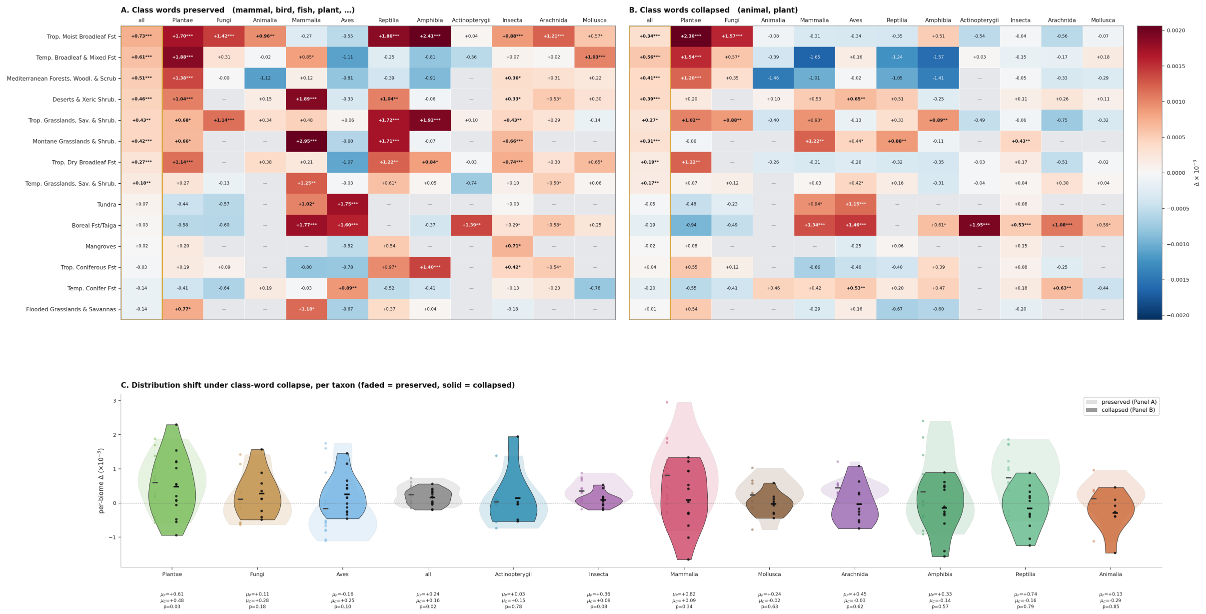

Figure 7. Per-taxon decomposition with class-word collapse control. Biome $\times$ iconic-taxon residualised Δ on sentence-pooled SigLIP-2 over the full LLM-clean corpus. *Panel A:* class words preserved – every specific species name has been replaced by its class word (`mammal` / `bird` / `fish` / `tree` / `plant`, etc.) but the class words remain distinct. *Panel B:* class words collapsed – every animal-kingdom class word further reduced to the single token `animal` and every plant-kingdom word to `plant`, so class-word frequency is identical across biomes by construction. Rows are the 14 WWF biomes, ordered by descending pooled (`all`) Δ in Panel A. Cell values are per-biome per-taxon Δ in $\times 10^{-3}$, with stars for nominal significance and bold for BH-$q < 0.05$ within the taxon. Cells with fewer than 10 biome–taxon photographs are shown as grey em-dashes. The gold-bordered `all` column anchors the marginal headline. *Panel C:* for each taxon, a double-violin contrasts the distribution of its 14 biome cells under the two encodings – faded behind for Panel A, solid in front for Panel B. Taxa are ordered left-to-right by descending collapsed mean μ_C. Mean markers and per-taxon Wilcoxon p-values (against $\Delta = 0$) appear beneath. Plantae, Fungi, Aves, and Insecta retain most of their across-biome alignment under collapse, indicating that the per-(biome, taxon) coupling for those taxa is carried by surrounding lexical-thematic content (activities, landform vocabulary, narrative shape) rather than by the class word alone. Mammalia, Reptilia, and Amphibia attenuate substantially, indicating that a large part of their preserved- encoding signal was class-word frequency mirroring image-class frequency.

Broadleaf, Mediterranean Forests on Plantae and Fungi; Boreal Forests/Taiga and Tropical Coniferous Forests on Insecta and Mollusca) tell us where the coupling is carried by content beyond the class word.

The collapse therefore partitions the per-taxon alignment into two components. For animal classes, the class word itself accounts for most of the per-cell work, and the residue after collapse is small. For plant and fungus classes, the alignment was carried from the start by something other than the class word, and that something survives the collapse largely intact. The Kruskal–Wallis omnibus across taxa under either encoding is not significant, and no individual taxon survives BH correction at the taxon level; with only 14 biome cells per taxon, the test has limited power to discriminate distributions across taxa and we do not claim a statistical separation between them.

Three components of the biome–mythology alignment can therefore be distinguished on the basis of the collapse. First, a weaker class-frequency ecological signal in which biome-b's text mentions a given class word more often because biome-b mythology is built around that class; the encoder reads the overlap from the names alone. This component is concentrated in the vertebrate animal taxa (Mammalia, Reptilia, Amphibia) and dissolves under the collapse. Second, a stronger residual lexical-thematic signal carried by the activities and narrative context the motif text builds around its biome's flora and fauna; this component is concentrated in the plant- and fungus-coded cells and survives the collapse largely intact. Third, the broader biome-level signal reported in Figure 3, which combines both contributions because the stratified per-biome Δ is the uniform mean of the per-(biome, taxon) cells from which both components originate. The conservative version of the biome-level coupling claim therefore runs through the second component, the part that survives the class-word collapse. We unpack what this asymmetry suggests about how mythology engages different parts of the biome in the Discussion.

Biome structure is recoverable without supervision

The analyses so far compare each motif to images grouped by a biome label we supply. As a converging check that never uses biome labels to build the representation, we re-described every motif by its *affinity profile*: the vector of its residualised cosine similarities to all 46,481 iNaturalist photographs (per-motif centred, as in the headline Δ). Two motifs are neighbours in this space when the same images draw them in and push them away, whatever those images depict. An unsupervised embedding of the geometry (PCA to 50 dimensions, then UMAP [McInnes et al., 2018]) is organised first by depicted *content*: colouring the map by each motif's dominant taxon recovers more structure (adjusted Rand index 0.08) than colouring it by cultural macro-area (0.04) or by biome (0.02). The geometry is dominated by what the myths are about rather than where they are from, and a biome-coloured scatter shows no clean clustering.

Biome is nonetheless present as a subdominant but reliable overlay, recoverable two ways without ever using biome labels to construct the affinity vectors (Figure 8). First, for each motif we ranked the nine well-sampled biomes by the motif's taxon-stratified mean similarity to their images and asked where the motif's own biome fell. Own biome lands at the 64th percentile on average against a 50th-percentile chance baseline ($p < .001$, 2,000 label permutations; Figure 8A). Second, a five-fold one-versus-rest logistic probe on the PCA-reduced affinity vectors decodes each biome above its label-shuffled null, at a mean balanced accuracy of 0.66 against 0.50 (Figure 8B). The affinity vectors are built from the anonymised text, in which species names are already class words, and both diagnostics are essentially unchanged when the text is collapsed further so that every animal class reads only "animal" and every plant class only "plant" (retrieval percentile 0.63, decodability 0.66 against a 0.50 null), so the recovered biome structure is not the species or taxon channel but the lexical-thematic content around it. The biome signal is real and recoverable from the model's image affinities alone, but it is a thin overlay on a geometry that is otherwise organised by the kinds of plants, animals, and scenes the myths describe.

The same recovery succeeds on the scene corpus. Repeating both diagnostics on the 1,655 Places365 landscape photographs — marginal rather than taxon-stratified, since the scenes carry no iconic-taxon labels — recovers biome from the myth–scene affinity geometry just as cleanly: across the seven biomes with sufficient scene and motif support, own biome lands at the 61st percentile ($p = 0.0005$, 2,000 permutations) and all seven decode above their label-shuffled null at Benjamini–Hochberg FDR ($q < .05$; mean balanced accuracy 0.68 versus 0.50; Figure 9). That the unsupervised geometry yields biome on two image corpora with no shared photographs and no shared visual content type — close-up species portraits versus wide-shot terrain — indicates the recovered structure is a property of the myths rather than of any one image distribution.

Discussion

The mythologies of nearly a thousand cultures, stripped of every species name, place name, biome word, and ethnonym we could identify, still align preferentially with images of the biomes those cultures inhabit. The alignment holds on two independent visual corpora, replicates across four vision–language models, and decays monotonically as motifs propagate beyond their biome of origin: tightest where the mythology is biome-specific, fading to nearly nothing where it has spread to most of the world. We read this pattern as evidence that mythology carries, in the lexical-thematic structure of its surviving text, a recoverable residue of the perceptual encounter between culture and biome.

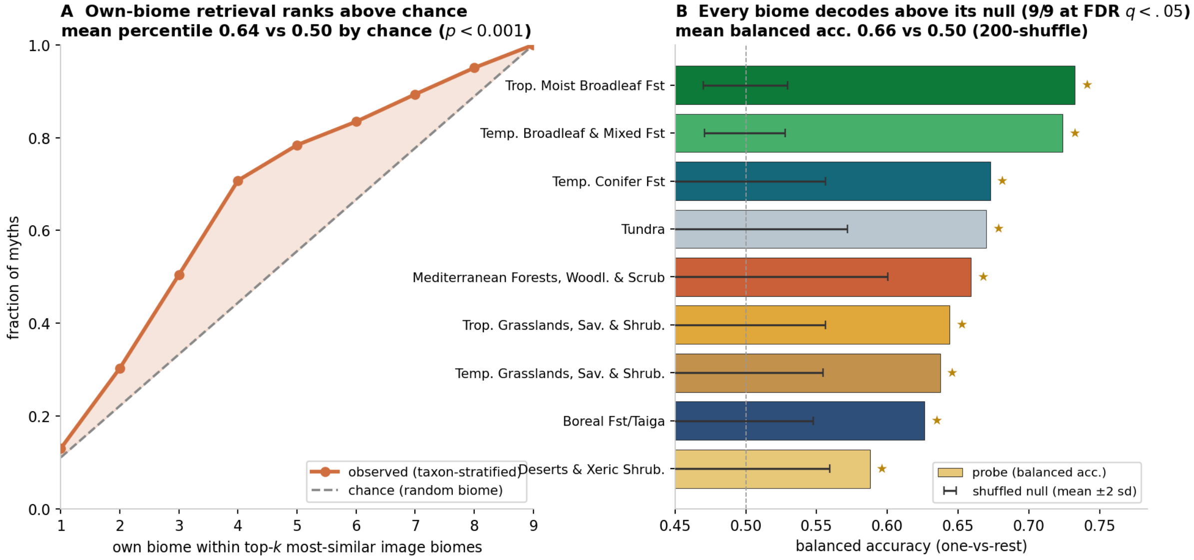

Figure 8. Biome structure is recoverable from the unsupervised myth–image affinity geometry, without using biome labels. Each motif is represented by its residualised cosine-similarity profile across all 46,481 iNaturalist photographs. **(A)** Own-biome retrieval. For each motif the nine well-sampled biomes are ranked by the motif's taxon-stratified mean similarity to their images; the curve gives the fraction of motifs whose own biome falls within the top k ranks (orange) against the chance diagonal (grey). Own biome reaches the 64th percentile on average ($p < .001$, 2,000 permutations). **(B)** Per-biome decodability. Balanced accuracy of a five-fold one-versus-rest logistic probe on the PCA-50 affinity vectors (bars) against a 200-shuffle label-permuted null (mean ± 2 sd); all nine biomes decode above chance at Benjamini–Hochberg FDR $q < .05$, at a mean of 0.66 versus 0.50. Throughout, "own biome" is the biome contributing the most of a motif's traditions, and the affinity vectors are built without reference to it.

Mythology as a recording layer of perceptual coupling

The breadth gradient is the result that most informatively distinguishes the two accounts of mythology we set out to test. A view in which mythology's symbolic content is largely a function of mind and culture and the environment supplies raw material but not form has no principled reason to expect that biome-specific motifs and universal motifs should align with their biomes' imagery at different rates. The two classes of motif are equally available as projection surfaces under that reading. A coupling reading accommodates the gradient more naturally. Biome-specific motifs are, on this account, the stable points of an encounter between a culture and a particular ecology, and their lexical-thematic content was shaped in that encounter. Universal motifs propagated precisely because they detached from any single biome and stabilised on cross-cultural structures (family conflict, transformation, the journey to and from the dead) whose conditions for stabilisation are not ecological. Universals fade because the coupling has been factored out of them.

That reading converts mythology, in the technical sense, into a *recording layer*: not a description of the biome, and not a projection onto it, but a residue of the work the biome did in shaping which narrative completions were ecologically reinforced and which were not. The motif catalogues of computational mythography [Berezkin, 2024, d'Huy et al., 2018, Tehrani, 2013] are, on this view, partial transcripts of a process that runs across many human generations and is irreducible to either side of the perceiver–environment pair. The phylogenetic mythology program [Tehrani, 2013, d'Huy and Tehrani, 2016] has shown that motifs can be tracked across cultural genealogies as faithfully as genes; the breadth gradient we observe adds a complementary claim, that within those genealogies the motifs that stay tied to a single biome are the ones whose ecological substrate kept selecting for the same coupling, while the motifs that spread are the ones whose stabilisation was no longer biome-dependent.

These readings sit on a spectrum, and the present data do not span it. A pure projection account, in which biome supplies only interchangeable narrative material, is the one reading the breadth gradient

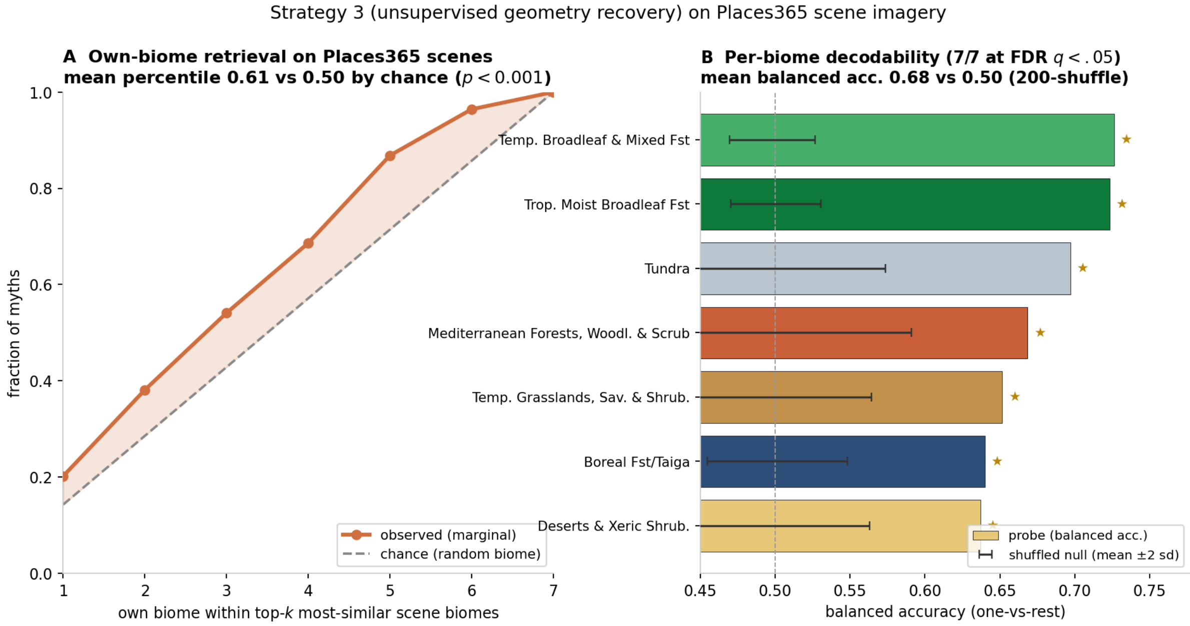

Figure 9. The unsupervised biome recovery replicates on Places365 scene imagery. As Figure 8, but each motif is represented by its similarity profile across the $1{,}655$ Places365 landscape photographs; because these scenes carry no iconic-taxon labels, own-biome retrieval uses the marginal per-biome similarity rather than the taxon-stratified one. **(A)** Over the seven biomes with sufficient scene and motif support, own biome reaches the 61st percentile on average ($p = 0.0005$, $2{,}000$ permutations). **(B)** All seven biomes decode above their 200-shuffle label-permuted null at Benjamini–Hochberg FDR $q < .05$, at a mean balanced accuracy of 0.68 versus 0.50. Biome labels are never used to build the affinity geometry.

makes uncomfortable. But between it and the strong coupling claim — that imagination is a perceptual organ jointly disclosing environmental form — lies a widely held middle position: that ecology constrains subsistence, attention, and the vocabulary a culture elaborates, and so the stories assembled from them, without imagination being anything so ambitious. Our measurement does not adjudicate between this middle position and the stronger coupling reading; both anticipate the ecological signature we find. It is therefore worth stating plainly what the data support, at three decreasing levels of strength. They *strongly* support that mythological texts carry ecological signatures beyond the explicit naming of species and places. They *more cautiously* support reading those signatures as the trace of a long-term perceptual and cultural coupling between communities and their environments. And they *do not decisively* establish that a coupling account explains the alignment where a cultural-projection account could not — an adjudication that would need the diachronic and cross-modal evidence sketched in the closing section.

A cognitive ecology of imagination

The lineage that makes this reading philosophically articulate was laid out in the Introduction — Schelling's *natura naturans*, Bateson's relocation of mind to wherever differences make a difference across coupled systems, the depth-psychological relocation of the image to the joint between psyche and nature, and Ingold's account of life as dwelling rather than projection [Schelling, 1797, 1799, Bateson, 1972, 1979, Jung, 1959, Hillman, 1975, 1981, Corbin, 1972, Abram, 1996, Ingold, 2000, Descola, 2013] — and need not be re-derived here. Its force, for the present result, is concrete: a human imagining a figure in a cliff face is not generating an inner image and applying it to outer matter but participating in a coupled system — neural pattern-completion machinery, the actual statistical structure of the cliff, the cultural priors about what counts as a figure, and the language that will or will not stabilise the percept into a transmissible name — in which pulling any element changes the event. The cognitive sciences supply the empirical scaffolding for reading imagination this way: Gibson's ecological psychology and the 4E program [Gibson, 1979, Varela et al., 1991, Clark and Chalmers, 1998, Noë, 2004], predictive processing as the mechanism by which a brain locks onto the interpretations its priors privilege given the world's actual statistical structure [Friston, 2010, Clark, 2016, Hohwy, 2013], and niche construction closing the loop, so that the symbolic environments humans build — names, stories, transmitted schemas — shape the cognition of later generations [Odling-Smee et al., 2003]. On this synthesis a figure in a biome is the stable point of a three-way coupling between environmental structure, perceptual machinery, and cultural transmission; imagination is not a private generative act but the moment of stabilisation in that coupling; and mythology is the cultural recording layer that keeps the stabilised completions available across generations.

That the breadth gradient favours the coupling reading over a projection one is the argument of the previous section; what the measurement adds is an unlikely detector. A vision–language model is a system without perceptual phenomenology, without cultural participation, and without priors shaped by deep evolutionary time. That such a system, trained on the public products of the coupling, recovers the coupling's signature in the joint statistical structure of environments and the stories told within them is what one would expect if the structure is really there to recover. It is, in Bateson's vocabulary, a difference that makes a difference at the joint, detectable from either side.

What kind of coupling: the lexical-thematic level

A coupling claim has to be carried at some level of the text. Our analyses locate it at the lexical-thematic level rather than at the level of narrative structure. The alignment survives word-shuffling of each motif, so sentence-level composition is not where the residue lives; it disappears on synthetic encyclopedic paragraphs, so the residualised statistic does not manufacture a positive on biome-neutral content (both in Supplementary Section S1, Figure 11); and it persists in motifs whose biome an independent, image-free text classifier cannot predict (a low-tell $\mu\Delta^{\text{strat}}$ of $+0.40 \times 10^{-3}$, essentially the full-corpus headline), so it is not reducible to a one-sided lexical artefact (Supplementary Section S4, Figure 14, Table 1). What is doing the work, in other words, is the distribution of generic landform vocabulary and biome-correlated activity schemas that the noun-targeted anonymisation pipeline retained by construction.

This level is the one a strict perceptual-coupling reading predicts. Activities (gathering, hunting, fishing, herding, fording, climbing) are not free parameters of culture overlaid on an inert environment; they are biome-constrained possibilities, the affordances Gibson named, that the environment makes available to the perceiver. Generic landform words (forest, mountain, river, snow, sand, sea) are not biome labels but the articulated joints of the perceived world, the structural features around which attention and narrative organise [Ingold, 2000, Descola, 2013]. A mythology that stabilises around the affordances and articulations of its biome is not failing to be symbolic; it is recording, in the only medium it has, the schemas of attention that the biome made available. The breadth gradient predicts that universals should be precisely the motifs whose stabilisation required affordances common to most environments, and that is what we find.

This is also the move that rescues the result from being merely methodological. The two-corpus replication is informative here. Activity vocabulary in the cleaned text could in principle align with activity *photographs* on a corpus that contained them. Places365's strict-landscape filter removes humans, vehicles, and depictions of activity, leaving only terrain and vegetation, and the alignment persists with comparable magnitude on that corpus. That is the signature of a coupling whose residue is read at the morphological level of the scene, not at the level of matching activity descriptions to activity images. The per-taxon decomposition makes a more careful version of the parallel argument from the species side, and what it shows is an asymmetry between the two kingdoms. The class-word collapse splits the per-(biome, taxon) signal into the part the encoder reads from the class word alone and the part carried by the surrounding lexical-thematic content. For animal-coded biomes the class word does most of the per-cell work. Boreal mythology references "mammal" more often because boreal mythology is full of bears and wolves; iNaturalist boreal photographs contain mammals because the boreal contains mammals; the encoder reads the overlap from the names alone. When the animal class words are collapsed to a single "animal" token, the Mammalia, Reptilia, and Amphibia cells deflate substantially because the alignment lived in the naming layer. For plant-coded and fungus-coded biomes the asymmetry reverses.

Plant naming in mythology is thin to begin with; most plants in the catalogue are already just "plant" or "tree" even before any further collapse, so the per-cell alignment had to come from something else from the start. The activities mythology built around plants (fruit-gathering, sacred groves, orchard-tending, vine-cultivation, the narrative roles plants take as origins, gifts, and transformations) are what align with the visual structure of biome-b's actual flora in their habitat, and they have nowhere else to go when the class word is removed. Plantae and Fungi cells survive the collapse with most of their magnitude intact across precisely the biomes where mythology engages local flora most thickly.

That asymmetry suggests two different modes of mythological engagement with the biome. Some inhabitants get named, walking through stories under their class identity — the mammals, reptiles, and amphibians whose vocabulary trace the encoder reads trivially. Others get encountered, narrated around, and stabilised through their roles in transformation, journey, and dwelling — the plants, fungi, birds, and insects whose mythological presence is built more from activities and contexts than from names. The class-word collapse separates these two modes and shows that both are real. The conservative version of the coupling claim runs through the second, where it can be most cleanly tested because the vocabulary that would carry a trivial alignment has been emptied out. That biome-b's plants and fungi still align with biome-b's mythology when the text says only "plant" is the cleanest single demonstration that mythology records something of its biome beyond the names of what lives there.

The anonymisation pipeline and the class-word collapse both make this point by *removing* naming from the text. The identity-naming analysis (Figure 5) reaches the same conclusion from the opposite direction, keeping the original myths intact and instead holding the species, place, and ethnonym channels constant statistically on the same stratified statistic. On that statistic the un-anonymised full myth aligns almost exactly as strongly as the anonymised one, and a bag of species names is more biome-diagnostic than the full myth, yet the alignment still persists in six of the fourteen biomes once all three naming channels are held constant, and in eight to twelve under a projection that removes the species axis from the embeddings. These two strategies rest on assumptions that do not overlap — one trusts the cleaning and removes the taxon confound, the other trusts no cleaning and removes the confound through stratification — and they converge on the same biomes. A third strategy reaches them from a third disjoint assumption: reading biome directly off the unsupervised myth–image affinity geometry, without ever using biome labels to build the representation, recovers each motif's own biome at the 64th percentile against a 50th-percentile baseline and decodes all nine well-sampled biomes above their label-shuffled null (Figure 8), and this recovery replicates on the Places365 scene corpus (Figure 9). Removing the names, holding them constant, and reading them off the geometry are three independent routes to the same biomes (Figure 1), and it is their convergence — methods with non-overlapping failure modes agreeing — rather than any single test that licenses reading the alignment as a property of the mythology rather than an artefact of how it was measured. The three routes do not all carry equally onto the scene corpus: there the conservative matched-permutation control retains only two of eleven biomes against six on the species channel (Supplementary Section S6), so we read that control as strongest on iNaturalist, while the anonymised alignment and the unsupervised recovery both hold on scenes as well. The biome–mythology alignment is, on all three readings, not reducible to the trivial channel by which a biome's mythology names that biome's species, places, and peoples.

Toward a falsifiable enactivist mythography

A philosophical reading that does not generate predictions is a worldview, not a research programme. The coupling lineage has historically been read as a worldview; the result reported here makes it possible to read it as a research programme with risky predictions. If imagination is coupling, and mythology is its recording layer, then several things should be true that follow naturally from this reading and are not naturally generated by a view in which biome supplies only interchangeable narrative material. They should be increasingly directly testable.

First, the same coupling signature should appear in mythological *visual art* more strongly than in mythological text, because art is a more direct transcript of the perceptual encounter and has not been compressed through the narrative-summary bottleneck Berezkin's catalogue records. Second, the signature should be denser in oral-narrative audio than in literary transcription, because audio preserves more of the contextual texture that text drops. Third, the breadth gradient should mirror the historical distribution of subsistence ecology: motifs that decoupled from biome should track the cultural transmission routes (trade, migration, religious conversion) that moved them, with their original ecological substrate fading as they entered other biomes. Fourth, motif-conditioned visual priors should improve scene classification on photographs of the biomes those motifs were stabilised in, because the priors encode information about which configurations were perceptually salient in that biome. None of these is a tautology; each can fail; and if mythology is doing what we suggest it is doing, each should pass.

The honest reading of what the data show is bounded. They do not demonstrate that nature is conscious, or that mythologies are messages from a non-human agency, or that the strongest metaphysical readings of the coupling lineage (Whitehead's process metaphysics [Whitehead, 1929], Goff's panpsychism or Strawson's realistic monism [Goff, 2019, Strawson, 2006]) are empirically supported. They show that the joint statistical structure of environments and the stories told within them carries a recoverable residue of their coupled history, robust enough that a system without phenomenology can detect it from either side. That is a smaller claim than the strongest readings of the lineage and a larger claim than a view in which biome supplies only interchangeable narrative material would allow.

Berezkin's corpus is curated, Eurasian-weighted, and sparse in several biomes (Mangroves, Flooded Grasslands, Temperate Conifer Forests). The vision–language models we use carry the biases of their training distributions, which the cross-model agreement bounds but does not erase. The anonymisation pipeline is noun-targeted; an activity-vocabulary pass that strips the remaining biome-correlated verbs is the natural next experiment, and we expect it to attenuate but not eliminate the signal, on the same logic that the within-Glottolog biome-swap null attenuated rather than eliminated it: the biome-content information is not concentrated in any single layer of the text but distributed across the substrate that the perceptual encounter shaped.

Schelling wrote that "nature should be visible Mind, and Mind invisible Nature." The pattern reported here is not a demonstration of his stronger thesis, but it is more naturally accommodated by a coupling or ecological-affordance reading of imagination than by a view in which biome supplies only interchangeable narrative material. Mythology, on the reading the data support, is one of the long instruments humans have used to keep the coupling articulate. The work of making the coupling reading testable has only begun.

Materials and Methods

Every analysis rests on one measurement — the alignment between a motif's sentence-pooled SigLIP-2 embedding and images of its biome — established three independent ways, the three strategies of Figure 1: removing the identity vocabulary from the text before embedding (*Controls against biome cues*), holding it constant on the raw myths (*Identity-naming controls*), and recovering biome from the unsupervised myth–image geometry without biome labels (*Unsupervised biome recovery*). The corpus, image sets, and embedding described next are shared across all three.

Corpus and biome assignment

We use Yu. E. Berezkin's *Analytical Catalogue of World Mythology and Folklore* [Berezkin, 2024], scraped from `ruthenia.ru` in May 2024. The catalogue provides 2,564 motif entries indexed across 958 traditions; each motif has a multi-paragraph Russian-language abstract aggregating the variants found across the traditions that carry the motif. Tradition centroids are joined to WWF Terrestrial Ecoregions [Olson et al., 2001] to assign each tradition, and through it each tradition–motif pair, to one of the 14 WWF biomes. Of the 2,564 catalogue motifs, the 2,158 analysed throughout are those with both a successfully scraped, anonymisable abstract and at least one tradition geocodable to a WWF biome; the remaining 406 lack one of these. Most of the excluded motifs have no biome-codable tradition and so cannot enter a biome-level statistic by construction, and the abstract-scrape failures are indexed by motif identifier rather than by biome, so the exclusion is not biome-targeted.

Image corpora

The iNaturalist component consists of $46{,}481$ images drawn from research-grade observations on iNaturalist Open Data [iNaturalist, 2024] across the 14 WWF biomes. We apply an image-side sanity filter that drops obvious non-nature content via SigLIP-2 zero-shot scoring against twelve negative prompts (*"a person in the photo"*, *"a portrait of a human"*, *"people holding something"*, *"an angler holding fish"*, *"an indoor scene"*, *"a kitchen or laboratory"*, *"a city skyline or street"*, *"a car or vehicle"*, *"a road sign"*, *"a black or all-white image"*, *"a museum specimen on a white background"*, *"a screenshot or document"*). An image is dropped if its maximum cosine similarity to any of the twelve negative prompts exceeds 0.08. The threshold is lenient (removing only obvious cases) so that species-in-habitat close-ups and natural outdoor scenes are retained, which is the content iNaturalist is biased toward by design. For the curated landscape comparison, the Places365 component [Zhou et al., 2018] is drawn from scene categories whose semantic interpretation unambiguously corresponds to a single WWF biome (e.g. *tundra* $\rightarrow$ arctic tundra; *rainforest* $\rightarrow$ tropical moist broadleaf; *forest/broadleaf* $\rightarrow$ temperate or tropical dry broadleaf; *snowfield/mountain_snowy* $\rightarrow$ boreal/taiga; *lagoon/swamp/creek/coast* $\rightarrow$ mangroves); categories that mix biomes were excluded. Each candidate image is then passed through a strict-landscape SigLIP-2 filter that drops anything that is not a wide-shot natural scene: five positive prompts (*"a wide-shot natural landscape with no people"*, *"a panoramic photograph of natural scenery"*, *"an empty natural habitat photographed from a distance"*, *"a landscape photo of natural terrain showing vegetation and horizon"*, *"a wild ecosystem with no human-made structures"*) and eleven negative prompts (*"a person in the photograph"*, *"a portrait of one or more humans"*, *"a close-up photo of a single animal"*, *"a single bird filling the frame"*, *"a car or vehicle"*, *"a boat"*, *"a road or street"*, *"a man-made structure or building"*, *"an indoor scene"*, *"a person riding a horse or animal"*, *"a tourist photograph with people"*) are encoded by SigLIP-2 and compared against each image's embedding. The strict-landscape score is mean(positives) $-$ max(negatives); an image is dropped if its max(negatives) > 0.18, and the remaining top-K images per biome are kept, yielding $1{,}655$ scene photographs across the 11 biomes Places365 covers. The Places365 filter is intentionally stricter than the iNaturalist filter, in particular it removes single-species close-ups, because the Places365 panel functions as a scene-only robustness check on biome-character imagery, complementary to iNaturalist's species-in-habitat coverage.

Vision–language embedding

The headline model is SigLIP-2-large (`google/siglip2-large-patch16-384`) [Tschannen et al., 2025], a multilingual vision–language model with native Russian support. Images use the standard 384×384 preprocessing. Text is sentence-pooled: each LLM-clean Berezkin abstract is split into sentences (NLTK Punkt sentence tokenizer [Kiss and Strunk, 2006], with a terminal-punctuation regex fallback; sentences of < 3 words dropped, capped at 50 sentences per motif), each sentence is embedded independently within SigLIP-2's 64-token window (mean 23 sentences per motif; $50{,}032$ sentences across all $2{,}158$ motifs), and the per-motif vector is the mean-pooled, L2-renormalised sentence embedding. Sentence pooling matters because the raw English abstracts average $\sim 3{,}400$ tokens, of which a 64-token truncation captures only the first $\sim 2\%$; the sentence-pooling pass embeds the full anonymised abstract instead. For cross-model robustness we additionally embed with M-CLIP (`XLM-Roberta-Large + CLIP-ViT-L/14`) [Carlsson et al., 2022] on the Russian abstracts (M-CLIP's 512-token window covers most abstracts in a single forward pass) and with OpenCLIP-ViT-L/14 under two pretrainings, LAION-2B [Schuhmann et al., 2022] and OpenAI [Radford et al., 2021], on English motif oneliners (see Supplementary Figure 13). Image and text features are L2-normalised and similarity is the inner product.

Controls against biome cues

The motif text fed to the encoder is anonymised against every category of biome-anchoring vocabulary we could identify, by an LLM-mediated translation and refinement pass followed by a single exclusion

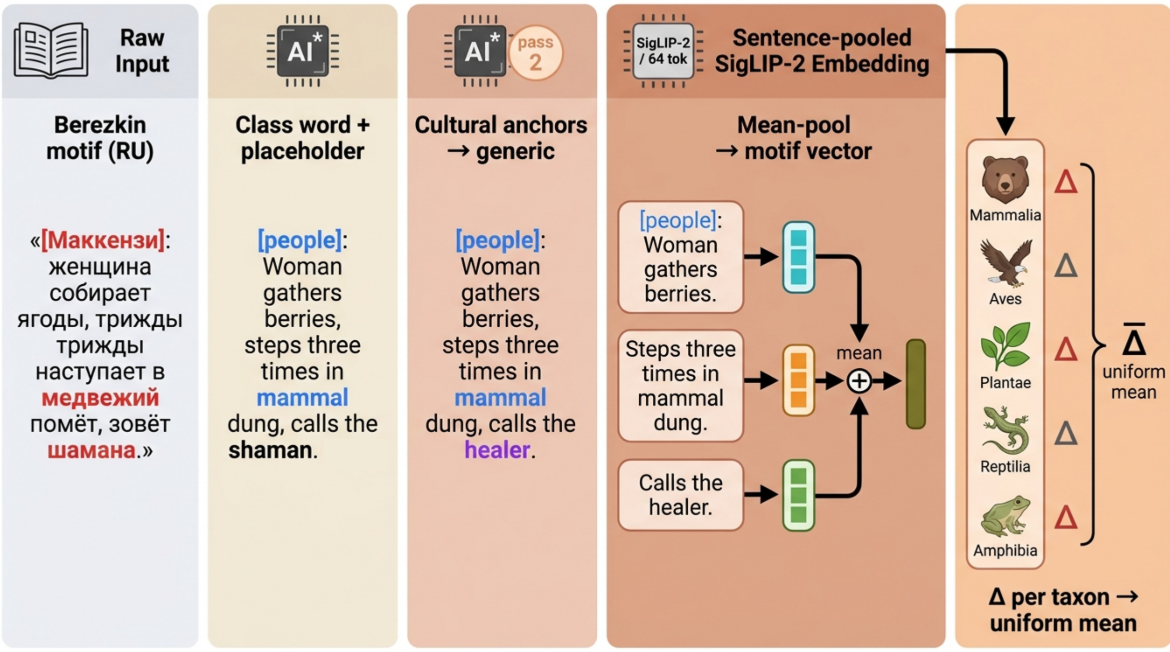

Figure 10. Pipeline schematic: anonymisation, sentence- pooled embedding, and image-side stratification. An example Berezkin motif (Mackenzie-region tradition, Boreal/Taiga) is threaded through the five stages of the analysis pipeline so the reader can trace one token's journey from raw Russian to the final per-taxon Δ. *Stage 0 (Raw input).* The Russian motif from Berezkin's catalogue, with biome-anchoring tokens highlighted in red (the species name "медвежий" bear, the cultural anchor "шамана" shaman, the ethnonym "Маккензи"). *Stage 1 (Class word + placeholder).* An LLM pass translates the text into English and applies seven noun-targeted anonymisation rules; specific species names become their class word (bear $\to$ `mammal`), ethnonyms become `[people]`, biome words are dropped. Replaced tokens are shown in blue. *Stage 2 (Cultural anchors $\to$ generic).* A second LLM pass refines long-tail cultural anchors that survived Stage 1 (`shaman` $\to$ `healer`, shown in purple); held-out audit on 200 random motifs confirms 96.5% strict-clean rate against species, place, and biome vocabulary. *Stage 3 (Sentence-pooled SigLIP-2 embedding).* The cleaned abstract is split into sentences, each sentence is embedded independently within SigLIP-2's 64-token window (mean 23 sentences per motif; 50,032 total across 2,158 motifs), and the per-motif vector is the mean-pooled, L2-renormalised sentence embedding. *Stage 4 (Δ per taxon $\to$ uniform mean).* On the image side, iNaturalist photographs of biome b are partitioned by iconic taxon (Mammalia, Aves, Plantae, Reptilia, Amphibia, …). The motif vector's residualised Δ is computed within each taxon stratum and averaged uniformly across taxa, removing biome-by-taxon co-occurrence on the image side as a possible explanation of alignment.

filter. For each motif we send the raw Russian Berezkin abstract together with the original English oneliner to gemini-3.5-flash (Google's frontier flash model, accessed 1 June 2026) [Google DeepMind, 2026] with a structured prompt that asks it to produce a faithful English translation under the following rules:

- Every specific species name $\to$ its class word: one of `mammal` / `bird` / `fish` / `reptile` / `amphibian` / `insect` / `mollusc` / `crustacean` / `arachnid` / `tree` / `flower` / `plant` / `fungus` / `animal`.
- Every country, region, river, mountain, lake, sea, ocean, or city name $\to$ the literal string `[place]`.
- Every tribe, ethnic group, nation, or ethno-religious community name $\to$ `[people]`.
- Every biome word (tundra, taiga, savanna, jungle, rainforest, prairie, desert, steppe, swamp, marsh, oasis, glacier, veldt, mangrove, mediterranean, etc.) dropped.
- Every cultivated crop (rice, corn, beans, peanuts, millet, quinoa, cassava, cotton, indigo, tobacco, etc.) $\to$ `plant`.

- Every biome-coded cultural artefact (sledge, kayak, harpoon, igloo, yurt, blowgun, snowshoes, kachina, calabash) $\to$ a generic equivalent (tool, boat, shelter, vessel).
- Latin binomials dropped.

A second Gemini pass applied to the first-pass output catches the long tail and additionally:

- Cultural-religious titles (sultan / padishah / shah / kadi / mullah / dervish / brahmin / bai / lama / raja / khan / tsar) $\to$ neutral "ruler" or "holy person" or "wealthy person" , explicitly *not* the Western-coded "king" / "queen" / "priest" / "monk".
- Tradition-specific named characters (Baba Yaga, Kivioq, Coyote-as-character, Nach Ak Yum, Quetzalcoatl, Khevki, Mitoshka, etc.) $\to$ "character".

Kept by design as cross-cultural anchors: cross-cultural religious figures (Christ, Buddha, Adam, Eve, Allah, God, Satan, named saints, Krishna, Rama, Sita), universal personifications (Sun, Moon, Wind, Sky, Earth, Sea, Thunder, Lightning, Star), all astronomical references (Pleiades, Orion, Big Dipper, Sirius, Venus, Milky Way, named constellations), all class words, cardinal directions, and generic universal vocabulary (forest, mountain, river, village, house, person, woman, man, child, day, night, fire, water). The narrative content of each motif, transformations, family relationships, magical events, plot, is preserved.

A held-out audit of 200 random LLM-anonymised motifs (criteria itemised in the Results panel C) finds 0% explicit biome words, 0.5% place or ethnonym residues, and 3.0% specific species residues, giving a composite strict-clean rate of 96.5% against species, place, and biome vocabulary. Generic landform vocabulary (forest, mountain, river, snow, sand, 92.5% of motifs) and biome-correlated activity verbs (gather, hunt, fish, herd, 93.5% of motifs) are retained by construction: the anonymisation pipeline is *noun-targeted*, stripping species, place, biome, and cultural-artefact nouns while leaving narrative structure and activity predicates intact. A residual narrative such as "gathers fruit, steps in mammal dung" therefore retains activity-level biome correlates (gathering fruit is more frequent in temperate and boreal forest descriptions than in desert ones). We treat this retained content as the substrate over which the alignment is measured rather than as contamination, and return to the morphology-versus-activity-priming question in the Discussion.

Within-taxon image-side stratification.

Because iNaturalist observations are taxonomically structured, some biomes are disproportionately rich in particular iconic taxa (boreal/tundra/conifer forests are 55–66% Plantae; mangroves only 11%). A motif whose anonymised text emphasises one taxon class will therefore score high in any biome whose iNat image pool emphasises the same taxon class, even when the biome's specific visual character contributes nothing. To remove this biome-by-taxon co-occurrence as a possible explanation for the alignment, we recompute Δ_b stratified by the iNaturalist iconic-taxon label. For each biome b and taxon stratum t with at least 20 images, we compute the within-stratum $\Delta_{b,t} = \overline{S_{i,\mathcal{M}_b}}_{i\in b\cap t} - \overline{S_{i,\mathcal{M}_{\bar{b}}}}_{i\in b\cap t}$, and report the uniform across-strata mean as $\Delta_b^{\text{strat}} = (1/T_b)\sum_t \Delta_{b,t}$. The permutation null is the same scrambled-motif null applied within strata. The headline iNaturalist statistic reported throughout the main text and main figures is stratified ($\mu\Delta^{\text{strat}} = +0.406 \times 10^{-3}$ across the 14 biomes), with the marginal $\mu\Delta^{\text{marg}} = +0.244 \times 10^{-3}$ reported alongside in robustness panels (where the input rather than the statistic is being varied) and on the Places365 panel of Figure 3 (Places365 has no iconic-taxon labels and therefore admits no further stratification). Per-text-taxon decompositions (Figure 7) report the per-cell marginal value at each biome–taxon cell, since each cell already conditions on a single taxon; the stratified Δ is, mechanically, the uniform mean of those per-cell marginals.

We separately explored simpler rule-based anonymisation pipelines (genus-level hypernymisation; a hand-curated class-level Russian hypernym map of roughly 3,500 entries; an audit-extended map of 6,300 entries) and found that the effect is positive under each of them and the across-biome ranking is preserved, ruling out the interpretation that any particular vocabulary anchor is doing the work. The LLM-mediated pipeline described above is the strictest in the family and the one we use for the headline.

Identity-naming controls

To test the alignment on the un-anonymised myths while controlling for biome-correlated naming, we extracted three identity classes per motif from the raw Russian Berezkin abstracts and kept them separate. Species (every specifically named organism, fauna and flora) were extracted by an LLM reading the Russian text; a held-out adversarial audit of 120 motifs estimated 96.7% recall and 95.3% precision against the source. Places are the WWF-codable regions in Berezkin's metadata, and ethnonyms are the tradition names that carry each motif. Each class was rendered as a bag of its entities, embedded with the same sentence-pooled SigLIP-2 pipeline as the full myth in the raw-Russian frame, and scored against the same iNaturalist images on the within-iconic- taxon stratified Δ. Because this is a self-contained raw-Russian decomposition, its absolute magnitudes are not formally comparable to the English LLM-clean headline, although the full-myth stratified Δ coincides closely with it. The matched-permutation null partitions motifs into 60 K-means blocks in the relevant identity-bag embedding space and, for each of $1{,}000$ permutations, shuffles biome labels only within a block, so identity content is held approximately constant while biome assignment is broken; the joint null concatenates the L2-normalised species, place, and ethnonym embeddings before blocking. Each biome is tested with an independent random-number stream, permutation p-values are $(1+\#\{null \geq obs\})/(1+1000)$ smoothed, and significance is reported after Benjamini–Hochberg FDR correction across the 14 biomes; the survivor counts in Figure 5B are at $q < .05$. The species-subspace projection removes the top k principal directions of the species-bag embeddings from the full-myth embeddings ($k = 5, 10, 20$) and recomputes the stratified Δ. The extraction prompts, audit, and per-biome tables are released with the code.

Statistical tests

Image–motif similarities form an $n_{\text{img}} \times n_{\text{motif}}$ matrix $\mathbf{S}$. Per-motif residualisation (subtracting the motif grand mean) eliminates the contribution of generic motif valence to per-biome aggregates. For each biome b with n_b images and motif partition $(\mathcal{M}_b, \mathcal{M}_{\bar{b}})$ where $\mathcal{M}_b$ is the set of motifs whose Berezkin tradition set includes at least one tradition assigned to biome b, we report

$$\Delta_b \;=\; \frac{1}{n_b} \sum_{i \in \text{imgs}(b)} \Big[\, \overline{S_{i,\mathcal{M}_b}} - \overline{S_{i,\mathcal{M}_{\bar{b}}}} \,\Big],$$

the residualised marginal Δ at the biome level. The headline iNat statistic is the within-iconic-taxon stratified version Δ_b^{strat} defined above. Significance is assessed by $1{,}000$ permutations of the motif-to-biome assignment (the stratified statistic recomputed under each permutation), with permutation $p = \#\{\text{null} \geq \text{obs}\}/1000$ and FDR control across biomes via the Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995]. The matched-permutation and unsupervised-recovery nulls below additionally use $+1$-smoothed p-values, $p = (1+\#\{\text{null} \geq \text{obs}\})/(1+n_{\text{perm}})$, and an independent random-number stream per biome. For all main-figure panels we apply a display threshold of $\geq$ 50 in-biome images, $\geq$ 5 in-biome motifs, and $\geq$ 20 images per biome–taxon stratum; biome–taxon cells with fewer than 10 images appear as missing in the per-taxon decomposition (Figure 7).

A fourth control, used in the cultural-geographic autocorrelation analysis, is the within-Glottolog-macroarea biome-swap null. Each tradition is assigned to one of the six Glottolog macro-areas [Hammarström et al., 2024] by a coordinate rule applied to its geographic centroid, rather than by a database lookup; under each of $1{,}000$ permutations every tradition's motifs are reassigned to a different biome that is also represented within its own macro-area, and the stratified Δ is recomputed. A biome is taken to carry signal beyond within-macro-area cultural regularity when its observed Δ exceeds the 95th percentile of this swap distribution (a one-sided 95th-percentile threshold; p-values are unadjusted across biomes). The null is defined once here and applied in the Results and in Supplementary Section S4.

Unsupervised biome recovery

To test whether biome structure is present in the model's image affinities without supervision, we represented each of the $2{,}158$ valid motifs by its row of the residualised similarity matrix $\mathbf{S}$, that is, its per-motif-centred cosine similarity to all $46{,}481$ iNaturalist photographs, and reduced these affinity vectors to 50 principal components. A motif's *own biome* is the biome contributing the most of its Berezkin traditions; this label is used only to evaluate the geometry, never to build it. We report two diagnostics over the nine biomes with at least 50 images and 20 own-biome motifs. First, own-biome retrieval: for each motif we computed a taxon-stratified mean similarity to each biome (the mean over iconic taxa of the motif's mean similarity to that biome's images of each taxon, requiring ≥ 20 images per biome–taxon cell), ranked the biomes, and recorded the rank of the motif's own biome; the chance baseline and the reported p-value come from $2{,}000$ permutations that reassign each motif's own biome at random. Second, per-biome decodability is the balanced accuracy of a five-fold one-versus-rest ℓ_2-regularised logistic probe on the PCA-reduced vectors, compared against a null of 200 such probes trained on permuted labels (each biome with an independent stream), with per-biome p-values smoothed as $(1 + \#\{null \geq obs\})/(1 + 200)$ and Benjamini–Hochberg FDR correction across the nine biomes. The unsupervised UMAP layout and its content, macro-area, and biome adjusted Rand indices use the same PCA-reduced vectors. As a text-side control against the taxon channel, both diagnostics are recomputed on the class-word-collapsed embeddings (every animal-kingdom class word mapped to "animal" and every plant-kingdom class word to "plant"; see *Controls against biome cues*). The same two diagnostics are repeated on the $1{,}655$ Places365 scenes; because the scenes carry no iconic-taxon labels, own-biome retrieval there uses the marginal rather than taxon-stratified similarity, and the PCA-50 is taken of the myth $\times$ Places365 similarity matrix, over the seven biomes with $\geq$ 50 scenes and $\geq$ 20 own-biome motifs (Figure 9).

Data and code availability

All code, sentence-pooled SigLIP-2 embeddings, the LLM-clean motif texts (Gemini-3.5-Flash output), the held-out anonymisation audit (the per-criterion coding of the 200 audited motifs is released as `anonymisation_audit_200motif.csv`), and the earlier rule-based hypernym maps released for completeness are available at [repository URL placeholder]. The Berezkin abstracts are accessible at http://www.ruthenia.ru/folklore/berezkin/.

Acknowledgements

[Placeholders.]

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Supplementary materials

The supplementary materials below collect the analyses that probe the robustness of the headline finding from six complementary angles: (i) the headline robustness atlas, a four-panel figure (word-shuffle, encyclopedic-register null, cross-corpus replication, and cross-model); (ii) a per-taxon decomposition that asks which iNaturalist taxonomic groups carry the alignment; (iii) a cross-model replication across four vision–language model architectures on the same LLM-clean text; (iv) two analyses directly addressing whether biome-correlated lexical content in the cleaned text could by itself explain the alignment (a high-tell versus low-tell split and a within-Glottolog-macroarea biome-swap null); (v) an analysis of the raw ecological composition of motif text per biome, which makes explicit what the anonymisation pipeline is removing; and (vi) a direct convergence check confirming that the two name-control strategies—removing the names and holding them constant—recover the same biome geography, breadth gradient, and per-taxon structure. Throughout, the same statistical conventions apply as in the main text: sentence-pooled SigLIP-2 on the full LLM-clean corpus, residualised Δ with $1{,}000$ motif-to-biome permutations and Benjamini–Hochberg FDR within each panel, and gold marks for $q < 0.05$.

S1. Robustness atlas

Four robustness analyses bound where the headline signal comes from. Word-shuffling each motif before sentence-pooling leaves the stratified $\mu\Delta$ essentially unchanged, indicating that the alignment depends on lexical content rather than narrative structure. Replacing the motifs with synthetic encyclopedic-register paragraphs of comparable length collapses $\mu\Delta$ to near zero, ruling out the possibility that the residualised statistic produces a positive bias on biome-neutral text. The per-biome stratified Δ on iNaturalist species photographs also tracks the marginal Δ on the entirely disjoint Places365 landscape corpus (Panel B; Pearson $r = +0.36$ across the 11 shared biomes, both means positive), an independent cross-channel replication on images that share no content type. Finally, the per-biome Δ replicates across all four vision–language model architectures we tested (SigLIP-2 sentence- pooled, M-CLIP, OpenCLIP-LAION-2B, OpenCLIP-OpenAI) with across-model Pearson correlations between $+0.74$ and $+0.94$, ruling out a SigLIP-specific training-distribution artefact (the full pairwise cross-model agreement is shown in S3, Figure 13). Finally, a held-out audit of 200 random anonymised motifs confirms a strict-clean rate of 96.5% against species, place, and biome vocabulary; the categories that remain by construction (generic landform words, activity verbs) are the substrate over which we measure alignment, and we interpret them as such in the main text Discussion.

S2. Per-taxon decomposition

Figure 12 re-renders the per-taxon decomposition of the headline result as small-multiple bar panels, one taxon per panel, with within-taxon BH-FDR. This is a complementary view of the same data as Figure 7: rather than collapsing a taxon's twelve biome cells into one violin, each panel shows the per-biome Δ values themselves, so the reader can see which specific biomes carry each taxon's effect. The "all" panel reproduces the headline ranking (Tropical Grasslands, Mediterranean, Boreal/Taiga at the top); the Aves and Plantae panels echo this pattern; the Reptilia, Mollusca, and Arachnida panels show little structure, consistent with the per-taxon violin summary.

S3. Cross-model replication

Figure 13 tests whether the per-biome stratified Δ pattern is an artefact of the specific vision–language model architecture used to produce the embeddings (sentence-pooled SigLIP-2-large in the headline analysis). For this robustness check we re-run the within-iconic- taxon stratified pipeline with four distinct vision–language model families on the same LLM-clean English motif text: sentence-pooled SigLIP-2-large; M-CLIP (XLM-Roberta-Large paired with CLIP-ViT-L/14, 512-token native window); OpenCLIP-ViT-L/14-LAION-2B; and OpenCLIP-ViT-L/14-OpenAI. The 4×4 scatter matrix shows per-biome stratified Δ correlations for every model pair; the off-diagonal cells report Pearson r. Correlations

Figure 11. Robustness atlas referenced in the main text. *Panel A*: per-biome stratified Δ on iNaturalist under three conditions that modify the text input. The original sentence-pooled embedding (dark blue) is essentially identical to its word-shuffled version (light blue) at every biome, indicating that the alignment depends on lexical content rather than narrative structure. The encyclopedic null (grey), in which synthetic encyclopedic-register English paragraphs of comparable length replace the LLM-clean motifs, collapses the alignment toward zero across biomes. *Panel B*: per-biome stratified Δ on iNaturalist species photographs (x-axis) versus marginal Δ on Places365 landscape scenes (y-axis), one point per WWF biome that both corpora cover. Pearson $r = +0.36$ across the 11 shared biomes; mean Δ positive on both sides ($+0.46 \times 10^{-3}$ iNat-shared stratified, $+0.42 \times 10^{-3}$ Places365 marginal). The two corpora share no images and no visual content type, so this is an independent cross-channel replication. *Panel C*: expanded 200-motif audit, per-criterion residue rate. Biome words, place or ethnonym residues, and specific species references are essentially eliminated (0%, 0.5%, and 3.0% respectively; composite strict-clean rate 96.5%). Generic landform vocabulary and biome-correlated activity verbs are retained by design. *Panel D*: cross-model headline stratified $\mu\Delta$ on the same LLM-clean motif text. Stratified $\mu\Delta$ is positive on all four architectures, from $+0.41 \times 10^{-3}$ on SigLIP-2 to $+0.74 \times 10^{-3}$ on OpenCLIP-LAION-2B.

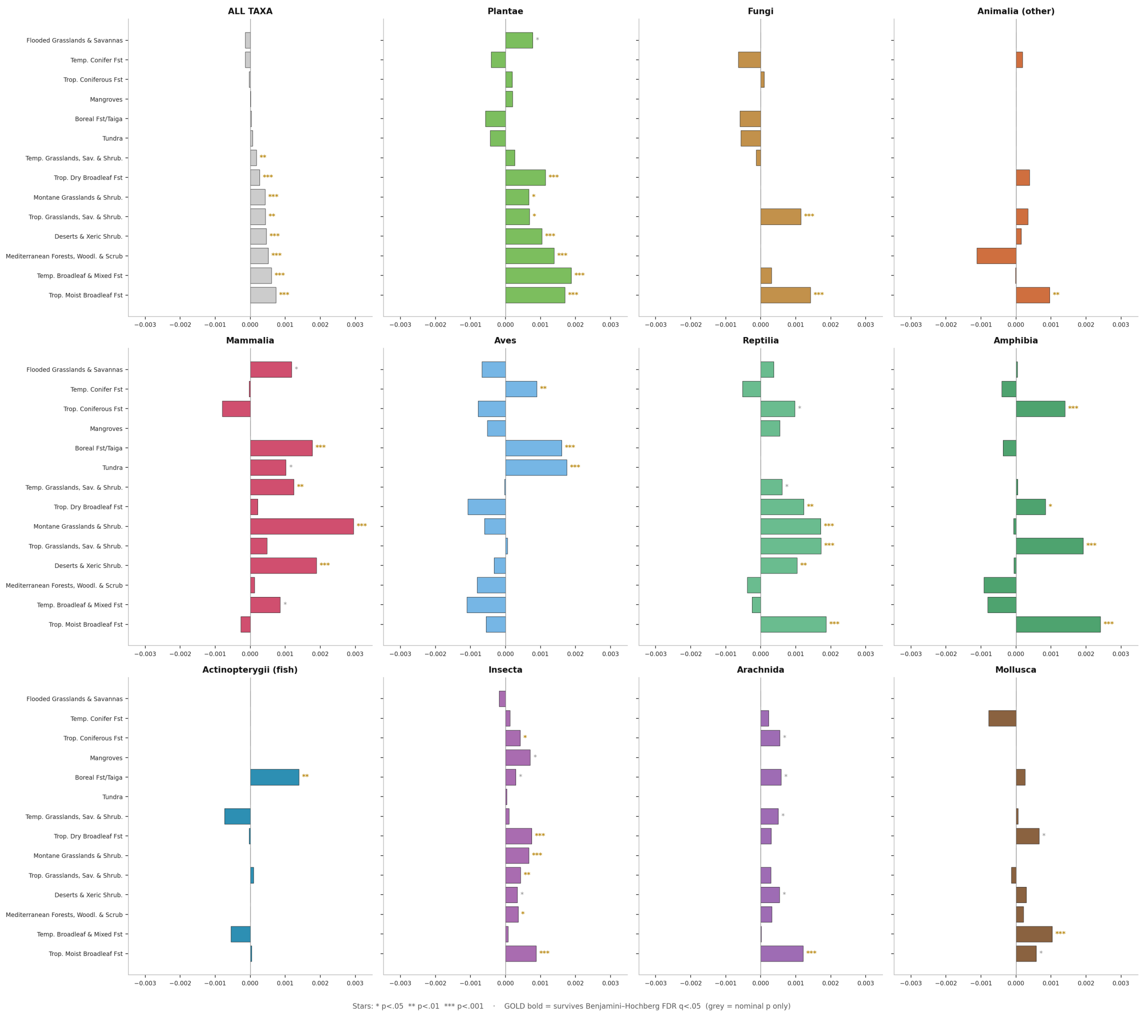

Figure 12. Per-taxon biome panels. Per-taxon decomposition of the headline result, presented as small-multiple bar panels with within-taxon BH-FDR. Each panel shows the per-biome Δ for one taxon; biome row order is preserved across panels. Gold stars mark BH-$q < 0.05$ within the panel. Complements Figure 7.

lie in $[+0.74, +0.94]$ across all six distinct model pairs, with the SigLIP-2-vs-OpenCLIP-LAION-2B pair carrying the strongest agreement ($r = +0.94$). The per-biome ranking is preserved across every model family tested, ruling out reading on which the headline pattern is a SigLIP-2-specific artefact of one model's training distribution.

S4. Biome-tell tests

A residual concern about the headline alignment is that the anonymised text might still contain enough biome-correlated vocabulary (generic landform words, activity verbs, class-level animal terms) for SigLIP-2 to recover biome assignment from text alone, with the image side being a secondary recovery of that lexical signal rather than an independent confirmation. We address this concern with two analyses whose details are provided in the data drop.

High-tell versus low-tell split. For each motif we compute a biome-tell z-score: the gap between its maximum sentence-pooled SigLIP-2 similarity to its assigned biomes' visual prototypes and the mean similarity to non-assigned biomes' prototypes, normalised by the standard deviation of off-target similarities. We split motifs at the median tell z-score and run the within-iconic-taxon stratified Δ test on each half (Table 1). The high-tell half (motifs whose anonymised text *is* biome- recoverable above chance) gives $\mu\Delta^{\text{strat}} = +0.54 \times 10^{-3}$ with 7 of 14 biomes significant at $p < .05$; the low-tell half (motifs

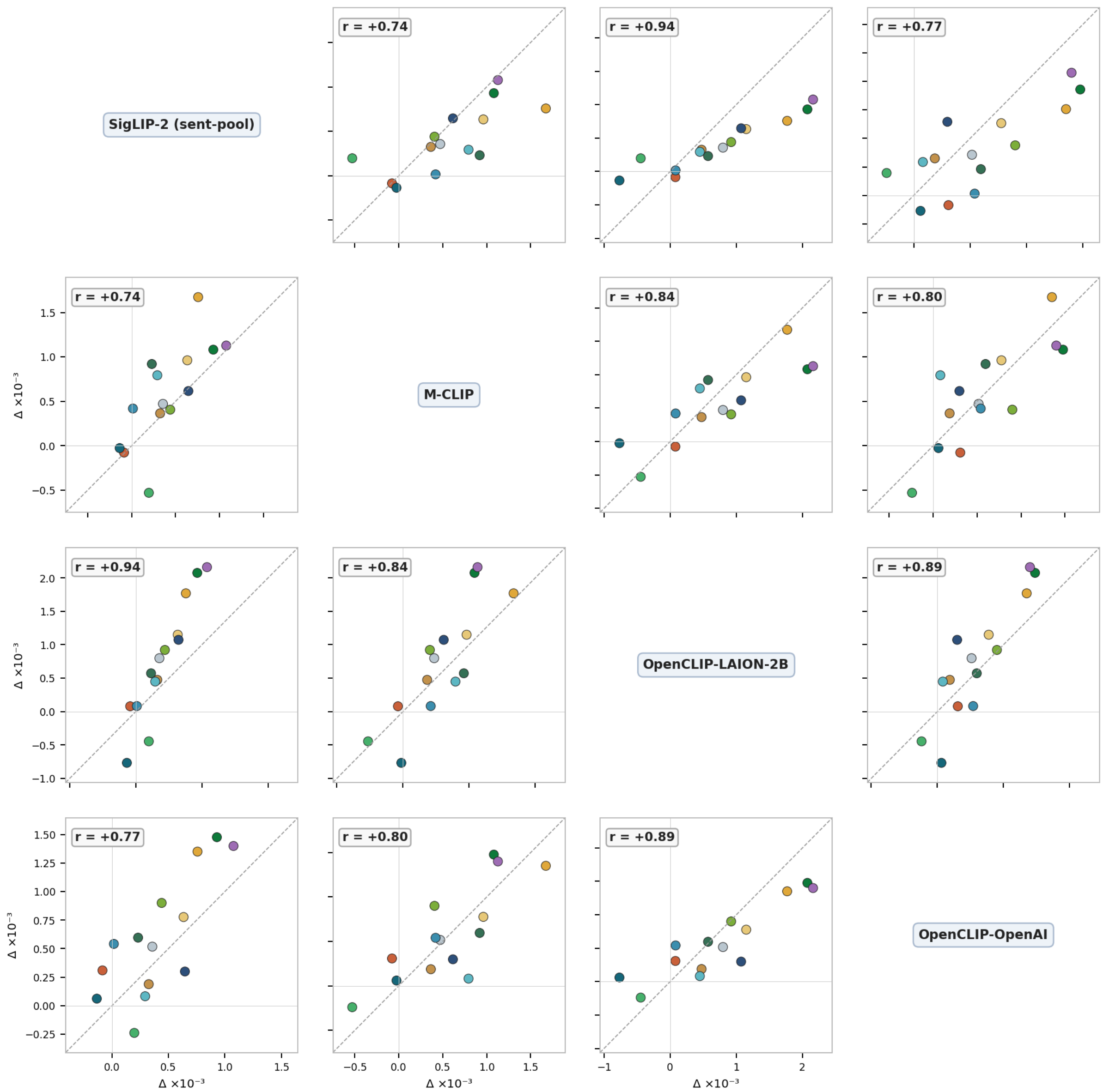

Figure 13. Cross-model robustness. 4×4 scatter matrix of per-biome within-iconic-taxon stratified Δ on the same LLM-clean Berezkin abstracts and the same iNaturalist images, across four vision–language model architectures. Off-diagonal cells report Pearson r between models; the dashed line marks $y = x$. Each point is one of the 14 WWF biomes, coloured by the standard biome palette. All six distinct model pairs correlate in $[+0.74, +0.94]$.

whose anonymised text alone is *least* biome-recoverable) still gives $\mu\Delta^{\text{strat}} = +0.21 \times 10^{-3}$ with 7 of 14 biomes significant at $p < .05$ (its marginal Δ is slightly negative, -0.41×10^{-3}, so the within-taxon stratification is what isolates the positive signal for this half). Text-only biome predictability therefore contributes to the alignment (the high-tell half is about $2.5\times$ the low-tell half), but the low-tell half on its own is not zero. That motifs whose text alone provides minimal biome predictability still align positively with their biome's imagery indicates that the headline signal is not fully explained by biome-correlated lexical content recoverable from text alone.

An image-free predictability split. The tell z-score above is computed from the vision model itself (the motif's SigLIP embedding against per-biome image prototypes), so the split variable and the alignment share SigLIP. As a fully independent check we replace it with a purely text-side predictor that never sees an image: a TF-IDF plus logistic-regression classifier trained to predict a motif's biome from its anonymised English text alone, evaluated out-of-fold (balanced accuracy 0.37 over the 10 well-populated biomes against a 0.10 chance baseline, so biome *is* partly predictable from the cleaned text). Splitting motifs at the median of this text-only predictability margin and recomputing the stratified Δ (Figure 14), the high-tell half gives $\mu\Delta^{\text{strat}} = +0.61 \times 10^{-3}$ (9 of 14 biomes significant) and the low-tell half — motifs whose biome a text classifier cannot guess — still gives $+0.40 \times 10^{-3}$ (6 of 14 significant), essentially the full-corpus headline of $+0.41 \times 10^{-3}$. Because this split is built with no reference to the image model, it rules out the reading on which the alignment is only the image side re-reading the same biome-correlated vocabulary a text model can already exploit.

Table 1. High-tell versus low-tell split under two predictability measures. Motifs are median-split by biome-tell and the within-iconic-taxon stratified Δ is recomputed on each half on iNaturalist images. The *vision* tell scores a motif by its SigLIP similarity to per-biome image prototypes; the *text-only* tell scores it by an out-of-fold TF-IDF plus logistic-regression biome classifier that never sees an image. Under both measures the low-tell half stays positive, and the text-only low-tell half is essentially the full-corpus headline.

Subset	$\mu\Delta^{\text{strat}}$ $(\times 10^{-3})$	$\mu\Delta^{\text{marg}}$ $(\times 10^{-3})$	Sig $p < .05$
Low-tell half (vision)	$+0.21$	-0.41	7/14
High-tell half (vision)	$+0.54$	$+0.78$	7/14
Low-tell half (text-only)	$+0.40$	$+0.29$	6/14
High-tell half (text-only)	$+0.61$	$+0.31$	9/14

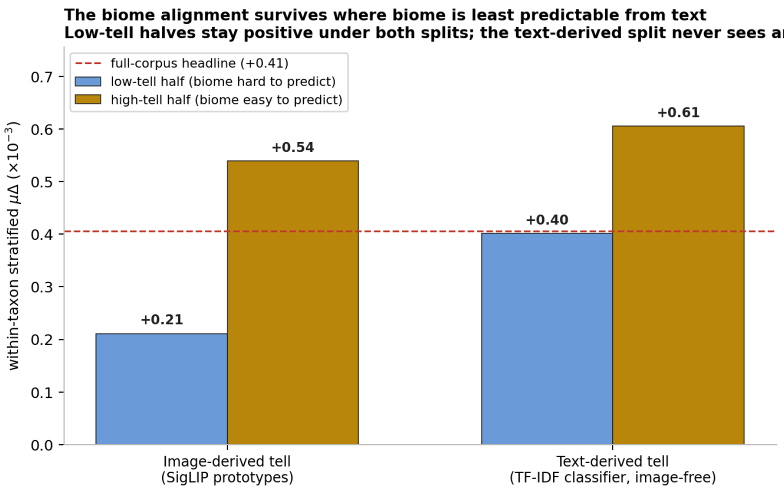

Figure 14. The biome alignment survives where biome is least predictable from the text alone. Within-iconic-taxon stratified $\mu\Delta$ on the low- and high-tell halves under two median splits: the vision-model tell (SigLIP myth-to-image-prototype similarity) and a fully image-free text-only tell (an out-of-fold TF-IDF plus logistic-regression biome classifier). Both low-tell halves stay positive; the text-only low-tell half reaches the full-corpus headline (dashed line, $+0.41 \times 10^{-3}$), so the alignment is not reducible to biome-correlated vocabulary a text model can exploit.

Within-Glottolog-macroarea biome-swap null. A stronger version of the macro-area block-permutation null is to actively swap biome labels while preserving cultural geography. For each tradition we randomly reassign its motifs to another tradition in the same Glottolog macro-area but a *different* biome, and recompute the per-biome stratified Δ under each of $1{,}000$ such swaps. If biome assignment is doing the work, the swapped Δ should collapse below the observed value; if biome identity is irrelevant and the alignment is purely a within-macro-area lexical artefact, the swap distribution should contain the observed value. Six biomes carry observed stratified Δ above the 95th percentile of the swap null distribution: Montane Grasslands & Shrublands (observed $+1.08 \times 10^{-3}$, swap-null mean $+0.09$, $p < .001$), Tropical & Subtropical Moist Broadleaf Forests (observed $+0.93$, null mean -0.16, $p < .001$), Boreal

Forests/Taiga (observed $+0.65$, null mean $+0.08$, $p < .001$), Deserts & Xeric Shrublands ($+0.63$ vs null $+0.22$, $p = .018$), Tropical & Subtropical Grasslands & Savannas ($+0.76$ vs null $+0.45$, $p = .019$), and Temperate Grasslands, Savannas & Shrublands ($+0.33$ vs null $+0.02$, $p = .022$). The remaining biomes either have observed values indistinguishable from the swap null (Tundra, Mediterranean, Temperate Conifer, Temperate Broadleaf) or are too sparsely represented for a stable swap distribution (Mangroves, Flooded Grasslands). This is a substantially stronger test than the standard motif-to-biome permutation, because it preserves within-macro-area cultural autocorrelation by construction: the biome label is being shuffled among traditions that share a linguistic-cultural region. That six biomes still carry signal above this stricter null indicates the alignment is not fully explained by within-macro-area lexical regularities — the biome content of the motif text itself carries information beyond what would be recoverable from cultural-region membership alone.

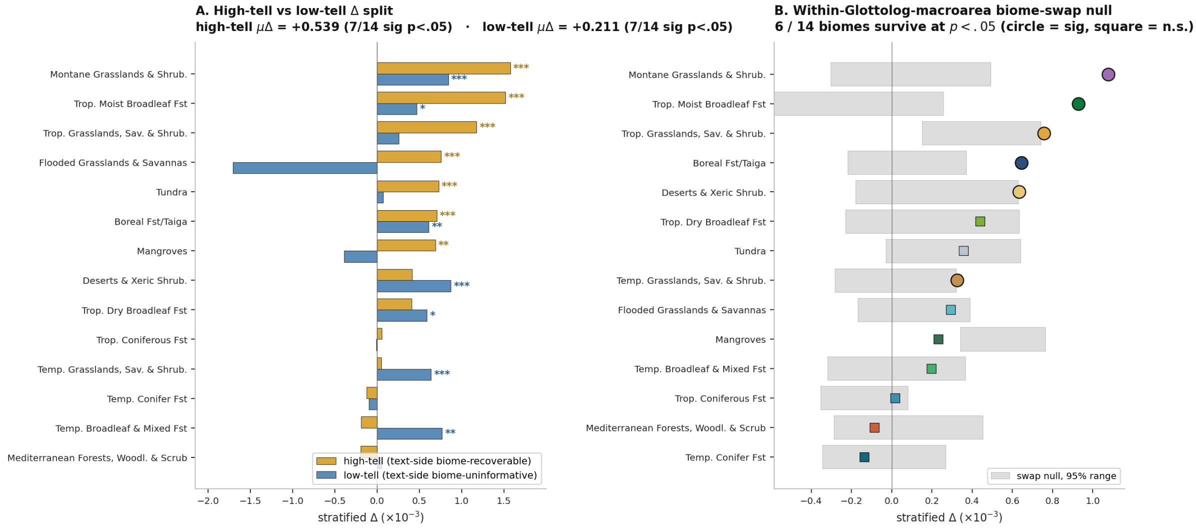

Figure 15. Biome-tell tests. *Panel A:* per-biome within-iconic-taxon stratified Δ on the high-tell half (motifs whose anonymised text is biome-recoverable above chance, gold bars) and the low-tell half (motifs whose text alone is biome-uninformative, blue bars), across the 14 WWF biomes. Both halves carry positive across-biome mean Δ ($+0.54$ and $+0.21 \times 10^{-3}$ respectively); seven biomes are individually significant at $p < .05$ in each half. *Panel B:* observed stratified Δ per biome (coloured markers) overlaid on the 95% range of the within-Glottolog-macroarea biome-swap null distribution (grey strips). Circles mark biomes whose observed Δ exceeds the 95th percentile of the swap null ($p < .05$); squares mark biomes that do not. Six biomes survive this stricter null, indicating that biome content carries information beyond within-macro-area cultural autocorrelation.

S5. Raw ecological composition per biome

Figure 16 reports the raw ecological composition of motif text per biome, before anonymisation. For each WWF biome (rows) and each of fourteen taxonomic and botanical categories (columns), the left panel of the heatmap shows the percentage of Spec A motifs (those touching ≤ 3 biomes, with ≥ 3 traditions in the biome) whose raw Russian abstract mentions at least one species or plant of that category. The right panel shows the difference between the Spec A subset and the all-in-biome pool: a positive cell means the category is over-represented in biome-specific motifs relative to the universal-dominated full pool. The heatmap therefore makes explicit, at the textual level, which taxonomic categories carry biome-specific signal before the hypernymisation layer removes their names from the encoder's input. The figure helps motivate which species-vocabulary categories the v7 hypernym map needed to cover comprehensively, and clarifies the ecological content the encoder is no longer reading at the headline stage.

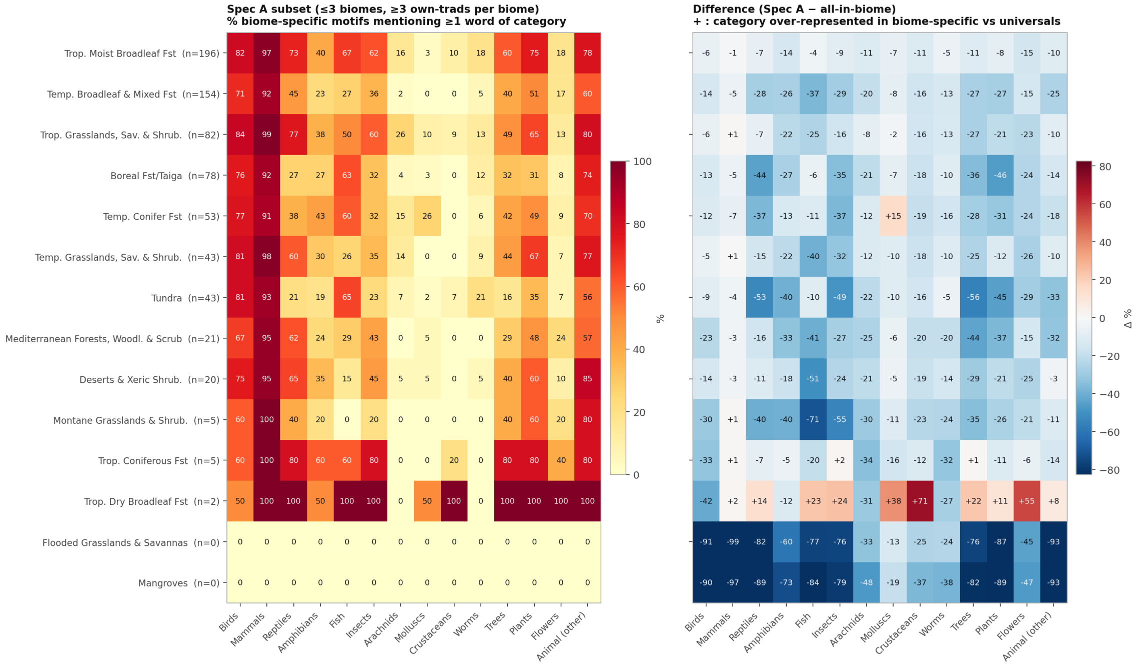

Figure 16. Ecological composition per biome (Spec A subset). *Left*: for each WWF biome (rows) and each of 14 taxonomic and botanical categories (columns), the cell shows the percentage of *Spec A motifs* (motifs touching ≤ 3 biomes, as in the breadth analysis) with at least three of their traditions in that biome, whose raw Russian abstract mentions at least one species or plant of that category, before anonymisation. *Right*: difference (Spec A minus all-in-biome). Positive cells mark categories over-represented in biome-specific motifs relative to the universal-dominated full pool, revealing the ecological fingerprint that the species-hypernymisation layer subsequently removes. The all-motifs version of the heatmap is dominated by universal motifs and therefore uniformly red across biomes; we omit it here and release the underlying counts in the data drop.

S6. Convergence of the two name-control strategies

The headline analyses control for identity naming by removing it from the text before embedding (the LLM-anonymisation pipeline). A complementary control leaves the raw myths intact and instead holds identity constant statistically: the matched-permutation null of the main text (Figure 5) shuffles the biome label only among motifs that share the same species, place, and ethnonym content—grouped into 60 K-means blocks of the concatenated, L2-normalised identity embeddings—and asks whether the observed within-iconic-taxon stratified Δ exceeds what identity alone determines. The two strategies rest on assumptions that do not overlap: one trusts the text anonymisation, the other trusts the identity matching and uses no cleaning at all. Placed side by side on iNaturalist species photographs, they recover the same biome–mythology coupling along the geography, breadth, and per-taxon axes the main text reports. (The paper's third strategy — unsupervised recovery of biome from the myth–image geometry without ever using biome labels — is reported in the main text and reaches the same biomes by a further independent route, on iNaturalist and on Places365 scenes alike; Figures 8–9.)

Geographically (Figure 17), the LLM-strip stratified Δ marks 8 of 14 biomes at Benjamini–Hochberg $q < 0.05$ and the raw-myth matched null marks 6 of 14, with five shared (Tropical & Subtropical Moist Broadleaf Forests, Boreal Forests/Taiga, Tropical & Subtropical Grasslands & Savannas, Montane Grasslands & Shrublands, and Tropical & Subtropical Dry Broadleaf Forests). The per-biome Δ values rank-correlate at Spearman $\rho = +0.75$ across all 14 biomes. The matched null is the more conservative of the two—it adds Mangroves but drops Deserts, Tundra, and Temperate Grasslands—and its survivor set is stable across the blocking resolution ($K = 30/60/120/240$ all give 5–6 of 14). The breadth gradient (Figure 18) replicates as well: both methods recover a monotone decay from biome-specific to universal motifs (the LLM-strip track here reproduces the main-text breadth gradient of Figure 6, overlaid against the raw matched-null track — LLM-strip $\mu\Delta^{\text{strat}} = +0.58 \to +0.44 \to +0.12 \times 10^{-3}$; raw matched null $+0.94 \rightarrow +0.42 \rightarrow +0.26$). And the per-taxon structure (Figure 19) lines up cell for cell: the per-(biome $\times$ taxon) Δ matrices correlate at $r = +0.64$ (49 of 113 cells nominally significant under the LLM-strip permutation null, 28 of 101 under the matched null with 16 surviving FDR), with the same vertebrate-and-plant taxa carrying the signal under both controls. We read this convergence as evidence that the alignment is a property of the motifs' lexical–thematic content rather than an artefact of any single anonymisation choice: it is equally present whether the identity names are physically removed from the text or held constant under permutation. The convergence does not extend uniformly to the Places365 scene corpus, where the matched null is markedly more conservative (two of eleven shared biomes at FDR versus six for the LLM-strip marginal Δ); we therefore read the matched null as a corroborating control that is strongest on the iNaturalist species channel.

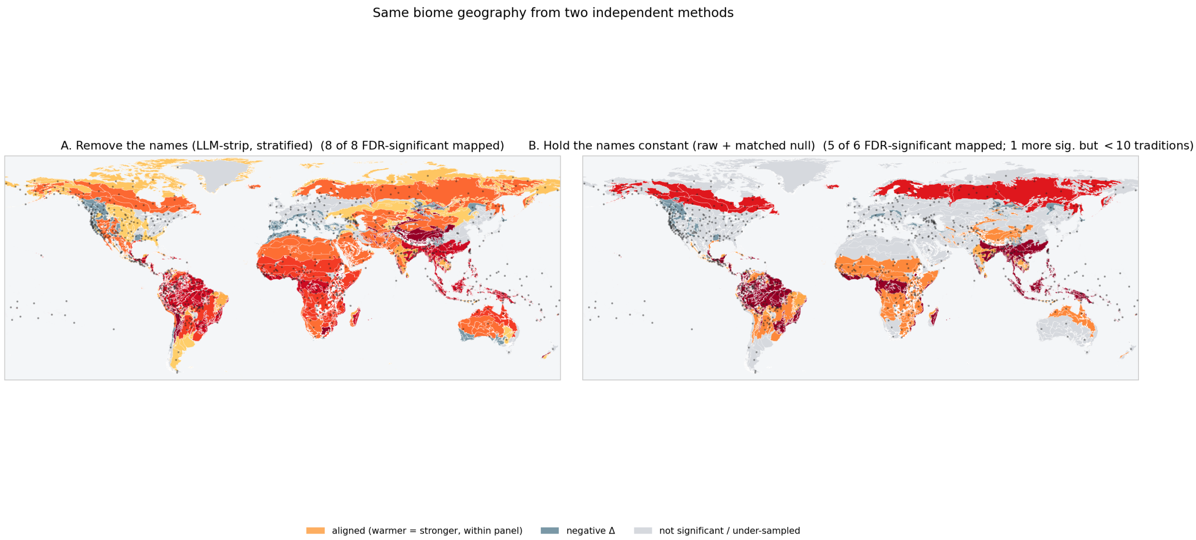

Figure 17. Same biome geography from two independent identity controls. Per-biome alignment on iNaturalist, mapped onto the WWF ecoregions. *Panel A* removes the names (LLM-strip within-iconic-taxon stratified Δ, motif-to-biome permutation null); *Panel B* holds the names constant (raw myths, discrete matched-permutation null). Each panel paints only the biomes that reach Benjamini–Hochberg $q < 0.05$ with positive Δ (8 of 14 and 6 of 14 respectively, five shared); one further matched-null biome (Mangroves) is significant but has fewer than ten traditions and is left grey. Warmer is stronger *within* each panel; the colour scales are per-panel and not comparable across panels. Black dots mark tradition centroids.

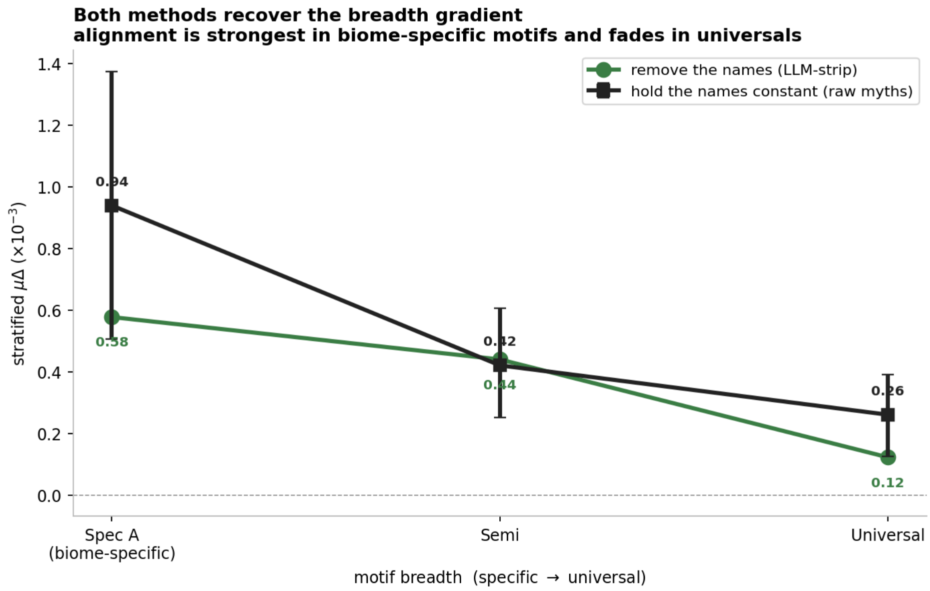

Figure 18. Both strategies recover the breadth gradient. Within-iconic-taxon stratified $\mu\Delta$ on iNaturalist by motif breadth (biome-specific Spec A $\rightarrow$ semi-universal $\rightarrow$ universal), for the LLM-strip corpus (green) and the raw myths under the matched null (black, with 95% bootstrap CIs). Both decay monotonically from specific to universal motifs. The raw track sits higher in magnitude because identity content is still present in the un-anonymised embedding, but the *shape* of the gradient is shared.

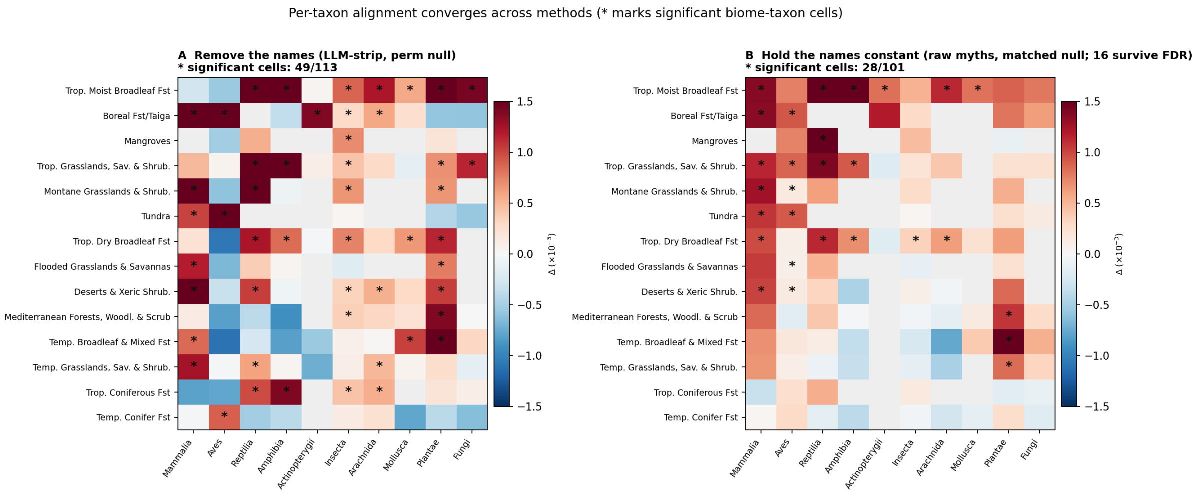

Figure 19. Per-taxon alignment converges across strategies. Per-(biome $\times$ iconic-taxon) Δ on iNaturalist. *Panel A*: remove the names (LLM-strip, motif-to-biome permutation null; 49 of 113 cells nominally significant). *Panel B*: hold the names constant (raw myths, matched null; 28 of 101 cells nominally significant, 16 surviving Benjamini–Hochberg FDR). Stars mark significant cells. The two matrices correlate at $r = +0.64$, and the same vertebrate-and-plant taxa carry the signal under both controls.